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Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis

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Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis

ABSTRACT

Introduction

Information technology can enhance timely and individual support for people with dementia and informal carers. There is some evidence that informal carers would benefit from technology-based counselling. However, it remains unclear which features of those interventions relate to beneficial outcomes and which aspects are necessary for a successful implementation. Therefore, the primary objectives are: (i) to identify conditions of successful implementation of technology-based counselling interventions in dementia and (ii) to investigate the effectiveness of those interventions.

Methods and analysis

We will conduct a mixed-methods systematic review. The first primary objective requires evidence from various study designs addressing aspects on effective and non-effective implementation of technology-based counselling. This could be telephone-, web- or mobile-based interventions for people with dementia and informal carers. For the second primary objective, randomised controlled trials (RCTs) dealing with any outcomes will be included. Publication years will not be restricted. We will search CINAHL, Cochrane Library, MEDLINE, PsycINFO, and Web of Science. Additionally, we will perform citation tracking of studies that met the inclusion criteria. To achieve the first primary objective, a Qualitative Comparative Analysis (QCA) will be conducted. The QCA enables us to identify necessary or sufficient components for a successful implementation. To reach the second primary objective, a meta-analysis will be performed with respect to potential clinical and statistical heterogeneity of RCTs. The revised Risk of Bias tool RoB 2.0 will be used to check the risk of bias in RCTs. For all other study designs, the Mixed Method Appraisal Tool will be used.

Ethics and dissemination

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Ethics approval is not required for this review. We will disseminate our findings through scientific and non-scientific journal articles and conference presentations as well as formats directed to the public and decision makers in health care.

Registration details

PROSPERO CRD42021245473

Strengths and limitations of this study

- A methodological strength of our review is the combined determination of effectiveness of technology-based counselling interventions and of successful implementation features of such interventions.
- A comprehensive literature search combining electronic key database search and supplementary search methods with no restrictions on publication date and outcomes.
- Qualitative Comparative Analysis might uncover effective and non-effective combination(s) of intervention components.
- Stakeholder participation throughout the review process enables us to take end-user perspectives.
- Probably few studies of technology-based counselling, especially computer-based counselling, could result in limited data for the Qualitative Comparative Analysis.

Keywords

Dementia, Technology-based counselling, Counselling, Implementation, Qualitative Comparative Analysis

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INTRODUCTION

Description of the condition

Dementia is one of the major causes of disability and dependency among older people and is characterized by a progressive decline of cognitive functioning alongside other impairments, such as memory, executive functions, attention, emotional control, and language.[1] Nearly 50 million people suffer from dementia worldwide, of whom approximately 60% live in low- and middle-income countries.[2] Over 9.9 million new cases of dementia are estimated each year worldwide.[3] The average survival time after diagnosis seems to depend on the type of dementia, however, the length of survival is not related to age at diagnosis.[4] Results of a meta-analysis demonstrate an average survival time of 4.11 years after the diagnosis of Lewy body dementia (SD±4.10). People with a diagnosis of Alzheimer's disease have an average survival time of 5.66 years (SD±5.32).[5]

Common consequences of dementia are behavioural and psychological symptoms. Hyperactivity and apathy show high persistence and incidence, depression and anxiety show low or moderate persistence and moderate incidence, and psychotic symptoms show low persistence and a moderate or low incidence.[6] Around half of all people with dementia need personal care. According to an estimation by Prince et al. informal carers spent an average of 14 hours weekly to support the person with dementia in activities of daily living (ADLs) such as clothing or bathing. Additionally, another 43 hours for instrumental activities of daily living (IADLs), for instance organisation of care or supervision are needed. No significant differences between countries were identified.[6]

The majority of people with dementia live at home and receive care by a relative, mainly a spouse.[7] Informal carers of people with dementia seem to be less healthy and might experience more difficulties and stress than informal carers of people with other chronic conditions.[8 9] Three dimensions of burden for informal carers have been described: direct impact of caregiving upon informal carers' life (e.g. lost control of life or health suffered), guilt (e.g. should do more or could do a better job), and frustration or embarrassment (e.g. care recipient asks for too much help or carer is embarrassed or angered by the care recipient).[10] Informal carers describe direct affects of caregiving such as anxiety, depression, loneliness, and stress, all of which influence sleep and cognition.[11] Nevertheless, informal caregiving does not only hold a negative connotation, because informal carers feel vindicated in their role or during the development of coping

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strategies.[12] Gender differences have been described in dealing with challenges of caregiving,[13] e.g. related to the point in time of seeking help [14] or the estimated availability of services.[15] Excessive informal carers' burden increases the risk of people with dementia having to move from their own home to a nursing home.[16] The worldwide COVID-19 (SARS-CoV-2) pandemic raised great concerns for people with dementia, e.g. due to a lack of information, difficulties in safeguarding procedures or social isolation.[17] Supporting those people and their informal carers is essential in order to overcome anxiety, stress, agitation or withdrawal.[18]

Description of the intervention

Information technologies provide supportive options without the risk of infection. Beyond the current pandemic, the intensity of dementia care and the lack of available support services or existing barriers to get access – especially in rural areas – require additional and alternative support possibilities.[19] Information technology can provide timely and individual support for people with dementia and informal carers.[20-26] Various types of support are needed, such as information, education or counselling.[27] Counselling is defined as a talking therapy that involves a trained therapist listening to individuals and helping them to find ways to cope with emotional issue. It is differentiated from information, education, instruction and case management.[28] Few studies have investigated the effects of counselling for carers of people with dementia.[29] Potential effects include changes in the attitudes of couples confronted with Alzheimer's disease towards becoming more acceptant of the diagnosis along with a more optimistic view for the future.[30]

How the intervention might work

Counselling can be provided through a face-to-face interaction, a group or technological channels, namely: the phone, e-mail, or live chat services. People who received telephone counselling believed that the intervention was attractive with regard to convenience, accessibility, control or inhibition.[31] A recent mixed-methods study uncovers the needs and preferences of informal carers of people with dementia and shows that an online intervention might be an adequate format to provide individual support. From their perspective, tailored programs that are simple to use and interaction with a professional within the online intervention are highly needed.[32] A recent scoping review on virtual support groups for informal carers of

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people with dementia also highlights the need for tailored, easy-accessible, and moderated online support.[33] Therefore, online counselling might meet the needs of many clients and results in positive outcomes.[34] Telephone counselling includes numerous aspects , such as psycho-education or behavioural activation.[35] Online counselling can be differentiated from other online services, e.g. web-based education or therapeutic interventions, therapeutic software or online activities, which are not stand-alone services.[36]

Alternatives to face-to-face interventions are crucial in the current COVID-19 pandemic and furthermore have the potential for a more timely and individual support in dementia. Few systematic reviews investigated technology-based interventions for informal carers of people with dementia, e.g. telephone-counselling,[37] or computer and telephone delivered support interventions.[35] The review by Lins et al. included nine randomised controlled trials (RCTs) and two qualitative studies.[37] The mixed-method review aimed to identify the efficacy of telephone counselling for informal carers of people with dementia and asked for experiences of informal carers with the intervention. The results demonstrate moderate evidence that telephone counselling without other intervention components can reduce depressive symptoms of informal carers. The experiences of informal carers included barriers and facilitators for successful implementation, counsellor's emotional attitude, and content of the telephone counselling. The authors concluded a lack of knowledge related to the aspects of telephone counselling, which influence the efficacy of the intervention. The review by Waller et al. included studies related to interventions delivered by a computer (n=10), multiple modalities (n=9), or telephone (n=15).[35] Due to the expanding number of related publications, one could assume an increasing demand for completing strategies for services, which have the potential to better support informal carers of people with dementia. In conclusion, a need for support and advice for the person with dementia and the informal carer as a dyad was identified.

An overview of systematic reviews investigated psychosocial interventions in dementia and identified counselling and psychotherapy as the second largest category of interventions for informal carers with some evidence to confirm the effectiveness of these interventions.[38] Technology-based interventions were also effective, but components or aims of these interventions were not described. The influence of several conditions of the effects of the interventions were not addressed in the overview. As well as in the reviews

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by Lins et al. and Waller et al., conditions of telephone-based counselling or technology-based interventions were not described.[35 37]

An update and extension of the review by Lins et al. is timely and warranted.[37] Additionally, to telephone-based counselling, we will cover other technologies such as computer or web-based interventions. Our review itself will address both, the target group of informal carers of people with dementia and studies that report on interventions for people with dementia themselves. That step would fulfil the global vision of evidence-based care in dementia, and to providing care and support for the need of people with dementia and their informal carers to enhance their dignity, respect, autonomy and equality.[39]

Objectives

Overall, there is some evidence that informal carers would benefit from technology-based counselling.[35 37] However, it remains unclear which features of these interventions relate to successful outcomes and which aspects of implementation are necessary and successful. The objectives of our review are (i) to identify conditions that are aligned with successful implementation of technology-based counselling interventions in dementia and (ii) to investigate the effectiveness of these interventions.

METHODS AND ANALYSIS

We will conduct a mixed-methods systematic review using (i) the methodological approach of Qualitative Comparative Analysis (QCA), aimed at identifying the features of or conditions for successful interventions,[40] and (ii) a meta-analysis, aimed at determining the overall effectiveness of technology-based counselling in dementia. PRISMA-P was used to structure this systematic review protocol.[41]

Eligibility criteria

To reach our review objectives, we are interested in every information concerning the development, evaluation, and implementation of technology-based counselling interventions in dementia. Therefore, we defined the following eligibility criteria.

Populations

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We will include studies on people with dementia and/or their informal carers living both at home and in skilled long-term care facilities. There will be no restrictions concerning gender, age, ethnic background, type and severity of dementia, and relationship between people with dementia and their informal carers. Studies on people with mild cognitive impairment or their informal carers will be excluded. Studies with a mixed sample of informal carers of people with dementia and other conditions will be included if the results for the dementia subgroup have been reported separately.

Interventions

We will include technology-based counselling interventions for people with dementia and/or their informal carers. Technology-based interventions use technologies such as telephone, computer, and web- or mobile-based applications to enable the remote delivery of the intervention. Counselling is a complex intervention that might consist of different intervention components. We will include counselling interventions practiced as talking therapy and provided by health care professionals such as nurses, physicians, or psychologists aiming to support people with dementia and/or their informal carers, e.g. focusing management of daily living, disease and care management as well as functioning and participation. Studies with interventions using counselling as a component of complex care programs such as case management will be excluded. We will also exclude specific interventions such as cognitive behaviour therapy, genetic counselling as well as counselling with regard to predictive diagnostics, screening for dementia or diagnostic procedures.

Comparators

Technology-based counselling will be compared to no treatment, to standard or usual care, and to in presence or face-to-face counselling. Comparisons between various technology-based interventions will also be considered for inclusion.

Outcomes

We will include all features of effective and non-effective intervention implementation and there will be no restrictions regarding outcomes on the effectiveness of interventions. Our primary outcome is defined as

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the impact of technology-based counselling on depressive symptoms experienced by carers of people with dementia. As secondary outcomes, we will consider outcomes on the (i) patient and carer levels (such as burden, well-being, delay or avoidance of transition to a nursing home, decision-making, and acceptance of as well as satisfaction with technology-based counselling) and on the (ii) healthcare system level (such as usage and costs of technology-based counselling).[35 38] We do not expect a negative impact from counselling interventions for people with dementia and carers. However, adverse events will be included.

Design of primary studies

The first primary objective requires evidence from various study designs addressing aspects of implementation success such as qualitative studies, descriptive quantitative studies (e.g. before-after studies, clinical controlled studies, cross-sectional studies, surveys), and mixed-methods research.

To meet the second primary objective, we will include individual and cluster parallel-group RCTs, also RCTs using cross-over and stepped-wedge designs.

Information sources

We will perform a comprehensive literature search combining electronic database search with supplementary search methods with no limitations regarding language and year of publication.[42 43]

Electronic database search

We will search CINAHL, Cochrane Library including CENTRAL (Cochrane Central Register of Controlled Trials), MEDLINE via PubMed, PsycINFO via Ovid, and the Web of Science Core Collection.

Supplementary search methods

Google Scholar will be used for free web searching with highly relevant free text terms to identify further studies and conference abstracts not yet identified as well as grey literature to minimise publication bias. We will search further primary studies using (i) hand searching of revealed reviews and (ii) forward and backward citation tracking of included studies using Scopus. To identify ongoing studies, we will search

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the International Clinical Trials Registry Platform (ICTRP), a meta study registry provided by the World Health Organization covering national registries from 17 countries worldwide.

Search update

An update for potentially relevant studies will be conducted in electronic databases one year after the initial search. For the search update, we will use the same search methods and strategies.

Search strategy

We will use database-specific search strategies based on database-specific controlled vocabulary such as MeSH (MEDLINE) and unspecified free text terms. The search strategies will be developed and peer reviewed within the review group using PRESS (Peer Review of Electronic Search Strategies). A researcher experienced in designing search strategies will support us during that process.[44] Our search strategy will contain three components (Dementia, Technology, and Counselling). Search terms and their synonyms will be identified through a previous work by our review group,[37] an orienting search using MEDLINE via PubMed, and the experience of the review group members. Additionally, text mining of titles and abstracts of previously known studies will be applied to identify relevant text terms used in the core literature. Based on the identified search terms, we will look for further search terms using a thesaurus [45] and checking entry terms given in the MeSH browser. For unspecified free text terms, we will search title, abstract, and keywords fields, if available. Controlled vocabulary will be used in the corresponding fields and additionally in the title, abstract, and keywords search fields in order to cover publications not yet tagged with controlled vocabulary.[46]

Study records

Two reviewers will independently screen the identified titles, abstracts, and, if eligible, full texts for inclusion and exclusion using the Rayyan web app.[47] We will discuss conflicting results within the screening team and with a third member of the review group - if necessary. Two reviewers will develop and pilot a data extraction sheet using ten percent of the included studies. Subsequently, one reviewer will

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extract data. A second reviewer will check data extraction for accuracy. We will discuss conflicting results within the data extracting team and, if necessary, also with a third member of the review group.

Data items

We will extract data on study characteristics (e.g. study design, country), participants' characteristics (e.g. age, gender, type and severity of dementia), intervention and control group characteristics (e.g. type, format, length, frequency), outcome(s) or phenomena of interest (e.g. implantation facilitators and barriers), measurement(s), and results. In addition, we will extract data according to the criteria of TIDieR and CReDECI-2 to provide a description of the reported intervention characteristics and their theoretical background.[48 49]

Outcomes and prioritization

We will involve our advisory board in determining the outcomes of interest following the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach (for details see below).[50]

Risk of bias and critical appraisal in individual studies

We will assess the risk of bias in RCTs using the revised risk of bias tool (RoB 2.0).[51] For all other study designs, we will use the Mixed Method Appraisal Tool, a comprehensive critical appraisal tool provided by Hong et al.[52] All appraisals will be conducted by two independent reviewers. We will discuss conflicting results within the appraisal team and, if necessary, with a third member of the review group.

Data synthesis

Qualitative Comparative Analysis (QCA)

To meet our first primary objective, we will conduct a QCA to identify conditions of interventions that are related to successful implementation and effective outcomes. Key elements of the QCA are the configurations of the conditions. These are different scenarios of intervention components and potential facilitators as well as barriers. The QCA enables us to identify necessary or sufficient components for

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implementation aspects and successful interventions. QCA differs from other analysis methods in not taking basic assumptions into account. QCA contains the steps illustrated in figure 1.[40]

INSERT FIGURE 1 HERE

Unlike systematic reviews on the effectiveness interventions, the QCA approach considers various study designs. Studies ("cases" in the QCA terminology) will be added or removed in an iterative process until a later stage of the research.

The selection and definition of the conditions and outcomes is based on the preliminary findings from the literature.[53] The conditions will be grouped into different domains such as setting and participants' characteristics, intervention characteristics, or planned and unplanned measures to facilitate implementation. The number of conditions is expected to be relatively low, and for each condition a clear hypothesis will be formulated according to its expected relation to the outcome.[40] Identified conditions and outcomes will be mapped in a raw data table and calibrated towards their so-called set membership. To prepare the calibration process, a coding framework is needed to provide instructions and values for the data extraction. For each case (= included study), the conditions and outcomes will be determined in crisp-sets with dichotomous values of 0 or 1, or with fuzzy-sets for degrees of membership, transformed in values between 0 and 1.

In a next step, the possible combinations of conditions will be tabulated. Each line in the table represents a configuration of possible conditions using logical operators. The number of configurations will be 2*k, where "k" represents the number of conditions. The configuration represents a potential set where studies can be members or non-members. Studies showing the same configuration will be included in a set and studies showing different configurations will be included in other sets. Therefore, so-called truth tables using the dichotomous or transformed values of crisp- and fuzzy-sets will be developed when configurations (i) are supported by (one or more) cases, (ii) are counter-intuitive or showed an identical combination of conditions, (iii) occurred infrequently, and (iv) triggered positive and negative outcomes at the same time.

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We will subsequently use Boolean minimisation to identify configurations that are observed to adequately lead to a sufficient and specific outcome. We will then consider the plausibility of the configurations and search for consistent conclusions in individual studies (consideration of logical remainders cases).

Identified sufficiently specific configurations can form different paths towards an outcome[53] Therefore, coefficients of coverage will be expressed, and the theoretical relevance will be considered. Furthermore, a "return to the cases" step will help us to evaluate how these results are represented in a study and across studies.[40 54]

The interpreted QCA results will be used for the development of a logical model that can support the practice of technology-based counselling. Furthermore, the newly introduced logical model has the potential to sufficiently inform further research activities in order to compromise inadequate knowledge about the effective steps of implementation. We will use an appropriate software for that analytic step, like TOSMANA or the QCA add-in for Excel.

Meta-Analysis

To meet our second primary objective, we will perform a meta-analysis with respect to the potential clinical and statistical heterogeneity of RCTs. Clinical heterogeneity will be indicated by the comparability of interventions based on intervention characteristics and components.[55] Groupings consisting of studies with similar characteristics and interventions might reach comparability. Therefore, it will be possible to conduct stratified meta-analyses given a sufficient number of studies across study groups. We will evaluate the statistical heterogeneity using the I^2 method.[56] A value higher than 50% will be regarded as significantly heterogeneous. Here, we will use the random-effects model and in the case of non-significant heterogeneity, we will use the fixed-effects model to calculate the pooled effect sizes.[57] A p value less than 0.05 will be considered as statistically significant. For meta-analysis of dichotomous results, we will calculate the relative risks (RR) with CI 95%. For the meta-analysis of continuous results, we will use the mean difference (MD) and CI 95%. If different outcome measurements were used, we will use the standardized mean difference (SMD) and CI 95%. To assess the influence of different conditions such as setting and recipient, delivery as well as type of the intervention on the effects of technology-based counselling, we

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will conduct sensitivity analyses based on subgroup meta-analyses.[58] We will use the software "R" in its current version for our calculations. The results of the meta-analysis will be presented in forest plots.[59]

If the previously described assessments indicate heterogeneous interventions and study results, we will provide a narrative synthesis based on tabular data extraction. Additionally, we will summarise effects, risk of bias, intervention and study characteristics graphically by means of a Harvest plot to report appropriately on the complexity and heterogeneity of the included studies.[60 61]

Meta-bias

Given a sufficient number of included studies to meet our second primary objective, risk for publication bias will be assessed graphically using a funnel plot complemented by a statistical test.[62]

Confidence in cumulative evidence

The quality of evidence will be assessed using the GRADE approach by rating the quality of evidence as high, moderate, low or very low.[63] The combined results of meta-analyses and the GRADE approach will be presented in a summary of findings table.[64]

Patient and Public Involvement

We will include representatives of people with dementia and informal carers, i.e. a person engaged in a support organisation that supports dementia care, and an informal carer of a person with dementia. The advisory board will consist of two or three people. Recommendations by the advisory board will be used for the qualitative steps of the QCA, where experience and knowledge of the subject are needed for the identification and the assessment of conditions that are important for the intervention. Furthermore, the advisory board will be involved in determining the outcomes of interest for the GRADE approach as well as the interpretation of the review findings.[50]

ETHICS AND DISSEMINATION

Ethics approval is not required for this review protocol. The members of the advisory board participate voluntarily in the review process. Ongoing informed consent will be obtained.

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Our dissemination strategy uses multiple channels of distribution to share our review work and results with academic and non-academic stakeholders, the society, and interested individuals.[65]

The final review will be published in an international open access journal relevant in the field of dementia care. Additionally, we will discuss our review results with experts at national and international conferences. Plain language summaries of the review will be placed on our institutional websites and social media profiles (e.g. Twitter, ResearchGate). A second publication will be published in a non-academic journal to share our review results with other interested groups such as practitioners and carers of people with dementia. Furthermore, we will participate in interactive mass media platforms to enable groups of interest to ask questions. A cartoon video will be accessible and disseminated online. Additionally, the results will be printed in newsletters of organisations representing people with dementia as well as patient advocacy groups such as Deutsche Alzheimer Gesellschaft, Alzheimer Schweiz, Österreichische Alzheimer Gesellschaft, and Alzheimer Europe. Furthermore, we will contact insurance companies and other stakeholders to disseminate our research results.

CONCLUSION

Although there is some evidence that informal carers would benefit from technology-based counselling, it remains unclear what features of those interventions relate to successful outcomes and what aspects of implementation are necessary and successful. Our review will identify conditions of a successful implementation of technology-based counselling interventions in dementia and investigate the effectiveness of those interventions. The results will show the effectiveness, i.e., benefits and potential adverse events of technology-based counselling for people with dementia and their carers. This is necessary to guide clinical practice concerning the use of such interventions. A comprehensive overview of successful and non-successful intervention components will facilitate a structured guidance for clinical implementation.

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Authors' contributions

GL and GM were involved in developing the idea for this review. All authors made substantial contributions to the conception and design of the work. JH and AB drafted the manuscript which was critically reviewed by all authors for important intellectual content. All authors gave their final approval of the manuscript version to be published and agreed to be accountable for all aspects of the manuscript.

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Competing interests' statement

All authors declare that there is no competing interest.

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REFERENCES

1. WHO. ICD-11 for Mortality and Morbidity Statistics. Neurocognitive disorders 2019 [Available from: https://www.who.int/classifications/icd/en/.
2. WHO. Dementia, 2019.
3. Prince M, Wimo A, Guerchet M, et al. World Alzheimer Report 2015: The Global Impact of Dementia: An Analysis of Prevalence, Incidence, Cost and Trend London2015 [Available from: https://www.alz.co.uk/research/WorldAlzheimerReport2015.pdf accessed 19.03.2020.
4. Brodaty H, Seeher K, Gibson L. Dementia time to death: a systematic literature review on survival time and years of life lost in people with dementia. *Int Psychogeriatr* 2012;24(7):1034-45. doi: 10.1017/S1041610211002924 [published Online First: 2012/02/14]
5. Mueller C, Soysal P, Rongve A, et al. Survival time and differences between dementia with Lewy bodies and Alzheimer's disease following diagnosis: A meta-analysis of longitudinal studies. *Ageing Res Rev* 2019;50:72-80. doi: 10.1016/j.arr.2019.01.005 [published Online First: 2019/01/10]
6. Prince M, Prina M, Guerchet M. World Alzheimer Report 2013: Journey of Caring: An analysis of long-term care for dementia London2013 [Available from: https://www.alz.co.uk/research/WorldAlzheimerReport2013.pdf accessed 19.03.2020.
7. Wimo A, Gauthier S, Prince M. Global estimates of informal care 2018 [Available from: https://www.alz.co.uk/adi/pdf/global-estimates-of-informal-care.pdf accessed 15.05.2020.
8. Schoenmakers B, Buntinx F, Delepeleire J. Factors determining the impact of care-giving on caregivers of elderly patients with dementia. A systematic literature review. *Maturitas* 2010;66(2):191-200. doi: 10.1016/j.maturitas.2010.02.009 [published Online First: 2010/03/24]
9. Isik AT, Soysal P, Solmi M, et al. Bidirectional relationship between caregiver burden and neuropsychiatric symptoms in patients with Alzheimer's disease: A narrative review. *Int J Geriatr Psychiatry* 2019;34(9):1326-34. doi: 10.1002/gps.4965 [published Online First: 2018/09/11]
10. Springate BA, Tremont G. Dimensions of caregiver burden in dementia: impact of demographic, mood, and care recipient variables. *Am J Geriatr Psychiatry* 2014;22(3):294-300. doi: 10.1016/j.jagp.2012.09.006 [published Online First: 2013/04/10]

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11. Bramboeck V, Moeller K, Marksteiner J, et al. Loneliness and Burden Perceived by Family Caregivers of Patients With Alzheimer Disease. *Am J Alzheimers Dis Other Demen* 2020;35:1533317520917788. doi: 10.1177/1533317520917788 [published Online First: 2020/04/14]

12. Lloyd J, Patterson T, Muers J. The positive aspects of caregiving in dementia: A critical review of the qualitative literature. *Dementia* 2016;15(6):1534-61. doi: 10.1177/1471301214564792 [published Online First: 2016/10/30]

13. Bieber A, Nguyen N, Meyer G, et al. Influences on the access to and use of formal community care by people with dementia and their informal caregivers: a scoping review. *BMC Health Serv Res* 2019;19(1):88. doi: 10.1186/s12913-018-3825-z [published Online First: 2019/02/03]

14. Brown J, Chen SL. Help-seeking patterns of older spousal caregivers of older adults with dementia. *Issues Ment Health Nurs* 2008;29(8):839-52. doi: 10.1080/01612840802182854 [published Online First: 2008/07/24]

15. Forbes DA, Jansen SL, Markle-Reide M, et al. Gender Differences in Use and Availability of Home and Community-Based Services for People with Dementia. *CJNR* 2008;40(1):38-59.

16. Cepoiu-Martin M, Tam-Tham H, Patten S, et al. Predictors of long-term care placement in persons with dementia: a systematic review and meta-analysis. *Int J Geriatr Psychiatry* 2016;31(11):1151-71. doi: 10.1002/gps.4449 [published Online First: 2016/10/19]

17. Wang H, Li T, Barbarino P, et al. Dementia care during COVID-19. *The Lancet* 2020;395(10231):1190-91. doi: 10.1016/s0140-6736(20)30755-8

18. WHO. ADI offers advice and support during COVID-19 https://www.alz.co.uk/news/adi-offers-advice-and-support-during-covid-192020 [Available from: https://www.alz.co.uk/news/adi-offers-advice-and-support-during-covid-19 accessed 05.05.2020.

19. Innes A, Morgan D, Kosteniuk J. Dementia care in rural and remote settings: a systematic review of informal/family caregiving. *Maturitas* 2011;68(1):34-46. doi: 10.1016/j.maturitas.2010.10.002 [published Online First: 2010/11/26]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20. Bejan A, Gündogdu R, Butz K, et al. Using multimedia information and communication technology (ICT) to provide added value to reminiscence therapy for people with dementia: Lessons learned from three field studies. *Z Gerontol Geriatr* 2017;51(1):9–15. doi: 10.1007/s00391-017-1347-7

21. D'Onofrio G, Sancarlo D, Ricciardi F, et al. Information and Communication Technologies for the Activities of Daily Living in Older Patients with Dementia: A Systematic Review. *J Alzheimer's Dis* 2017;Article in Press doi: 10.3233/JAD-161145

22. Lauriks S, Reinersmann A, van der Roest HG, et al. Review of ICT-based services for identified unmet needs in people with dementia. *Ageing Res Rev* 2007;6(3):223–46. doi: 10.1016/j.arr.2007.07.002

23. Lucero RJ, Fehlberg EA, Patel AGM, et al. The effects of information and communication technologies on informal caregivers of persons living with dementia: A systematic review. *Alzheimers Dement* 2019;5:1–12. doi: 10.1016/j.trci.2018.11.003

24. McHugh JE, Wherton JP, Prendergast DK, et al. Identifying opportunities for supporting caregivers of persons with dementia through information and communication technology. *Gerontechnology* 2013;10(4) doi: 10.4017/gt.2012.10.4.003.00

25. Noone C, McSharry J, Smalle M, et al. Video calls for reducing social isolation and loneliness in older people: a rapid review. *Cochrane Database Syst Rev* 2020;5:CD013632. doi: 10.1002/14651858.CD013632

26. Pleasant M, Molinari V, Dobbs D, et al. Effectiveness of online dementia caregivers training programs: A systematic review. *Geriatr Nurs* 2020;Article in Press doi: 10.1016/j.gerinurse.2020.07.004

27. Whitlatch CJ, Orsulic-Jeras S. Meeting the Informational, Educational, and Psychosocial Support Needs of Persons Living With Dementia and Their Family Caregivers. *Gerontologist* 2018;58(suppl_1):S58-S73. doi: 10.1093/geront/gnx162 [published Online First: 2018/01/24]

28. NHS. Counselling https://www.nhs.uk/conditions/counselling/2017 [accessed 17.1.2020.

29. Dickinson C, Dow J, Gibson G, et al. Psychosocial intervention for carers of people with dementia: What components are most effective and when? A systematic review of systematic reviews. *Int Psychogeriatr* 2017;29(1):31-43. doi: 10.1017/S1041610216001447 [published Online First: 2016/09/27]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

30. Elvish R, Lever S-J, Johnstone J, et al. Psychological interventions for carers of people with dementia: A systematic review of quantitative and qualitative evidence. *Counsell Psychother Res J* 2013;13(2):106-25. doi: 10.1080/14733145.2012.739632

31. Reese RJ, Conoley C, W., Brossart DF. The Attractiveness of Telephone Counseling: An Empirical Investigation of Client Perceptions. *J Couns Dev* 2006;84:54-60.

32. Teles S, Paúl C, Sosa Napolskij M, et al. Dementia caregivers training needs and preferences for online interventions: a mixed-methods study. *J Clin Nurs* 2020;Article in Press doi: 10.1111/jocn.15445

33. Armstrong MJ, Alliance S. Virtual Support Groups for Informal Caregivers of Individuals With Dementia: A Scoping Review. *Alzheimer Dis Assoc Disord* 2019;33(4):362–69. doi: 10.1097/WAD.0000000000000349

34. Harris B, Birnbaum R. Ethical and Legal Implications on the Use of Technology in Counselling. *Clin Soc Work J* 2015;43:133–41.

35. Waller A, Dilworth S, Mansfield E, et al. Computer and telephone delivered interventions to support caregivers of people with dementia: a systematic review of research output and quality. *BMC Geriatr* 2017;17(1):265. doi: 10.1186/s12877-017-0654-6 [published Online First: 2017/11/18]

36. Dowling M, Rickwood D. Online Counseling and Therapy for Mental Health Problems: A Systematic Review of Individual Synchronous Interventions Using Chat. *J Tech Hum Serv* 2013;31(1):1-21.

37. Lins S, Hayder-Beichel D, Ruecker G, et al. Efficacy and experiences of telephone counselling for informal carers of people with dementia. *Cochrane Database Syst Rev* 2014(9):CD009126. doi: 10.1002/14651858.CD009126.pub2

38. Clarkson P, Hughes J, Xie C, et al. Overview of systematic reviews: Effective home support in dementia care, components and impacts-Stage 1, psychosocial interventions for dementia. *J Adv Nurs* 2017;73(12):2845-63. doi: 10.1111/jan.13362

39. WHO. Global action plan on the public health response to dementia 2017 - 2025. 2017

40. Rihoux B, Ragin CC, editors. *Configurational comparative methods: Qualitative comparative analysis (QCA) and related techniques*. Los Angeles: SAGE, 2009.

41. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1. doi: 10.1186/2046-4053-4-1

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

42. Cooper C, Booth A, Varley-Campbell J, et al. Defining the process to literature searching in systematic reviews: a literature review of guidance and supporting studies. *BMC Med Res Methodol* 2018;18:85. doi: 10.1186/s12874-018-0545-3

43. Lefebvre C, Glanville J, Briscoe S, et al. Searching for and selecting studies. In: Higgins JPT, Thomas J, eds. Cochrane Handbook for Systematic Reviews of Interventions. 2 ed. Hoboken: Wiley Online Library 2019:67–108.

44. McGowan J, Sampson M, Salzwedel DM, et al. PRESS Peer Review of Electronic Search Strategies: 2015 Guideline Statement. *J Clin Epidemiol* 2016;75:40–46.

45. Dictionary.com. Thesaurus.com 2020 [Available from: http://www.thesaurus.com/ accessed 28.04.2020.

46. Rodriguez RW. Comparison of indexing times among articles from medical, nursing, and pharmacy journals. *Am J Health Syst Pharm* 2016;73(8):569–75. doi: 10.2146/ajhp150319

47. Ouzzani M, Hammady H, Fedorowicz Z, et al. Rayyan-a web and mobile app for systematic reviews. *Syst Rev* 2016;5:210. doi: 10.1186/s13643-016-0384-4

48. Hoffmann TC, Glasziou P, Boutron I, et al. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *BMJ* 2014;348:g1687.

49. Möhler R, Köpke S, Meyer G. Criteria for Reporting the Development and Evaluation of Complex Interventions in healthcare: revised guideline (CReDECI 2). *Trials* 2015;16(1):204.

50. Langer G, Meerpohl JJ, Perleth M, et al. GRADE-Leitlinien: 2. Formulierung der Fragestellung und Entscheidung über wichtige Endpunkte. *Z Evid Fortbild Qual Gesundhwes* 2012;106(5):369–76.

51. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;2:l4898. doi: 10.1136/bmj.l4898

52. Hong QN, Fàbregues S, Bartlett G, et al. The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. *Education for Information* 2018;34(4):285–91. doi: 10.3233/EFI-180221

53. Schneider CQ, Wagemann C. Standards of Good Practice in Qualitative Comparative Analysis (QCA) and Fuzzy-Sets. *Comp Sociol* 2010;9(3):397–418. doi: 10.1163/156913210X12493538729793

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

54. Kahwati L, Jacobs S, Kane H, et al. Using qualitative comparative analysis in a systematic review of a complex intervention. *Syst Rev* 2016;5:82. doi: 10.1186/s13643-016-0256-y

55. West SL, Gartlehner G, Mansfield AJ, et al. Comparative Effectiveness Review Methods: Clinical Heterogeneity. Agency for Healthcare Research and Quality. Rockville, 2010.

56. Higgins JPT, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557–60. doi: 10.1136/bmj.327.7414.557

57. Hedges LV, Vevea J. Fixed-and random-effects models in meta-analysis. *Psychol Methods* 1998;3(4):486–504.

58. Khan KS, Kunz R, Kleijnen J, et al. Systematic Reviews to Support Evidence Bsaed Medicine. How to Review and Apply Findings of Healthcare Research London: Hodder & Stoughton.

59. Lewis S, Clarke M. Forest plots: trying to see the wood and the trees. *BMJ* 2001;322(7300):1479–80. doi: 10.1136/bmj.322.7300.1479

60. Higgins JPT, López-López JA, Becker BJ, et al. Synthesising quantitative evidence in systematic reviews of complex health interventions. *BMJ Glob Health* 2019;4(Suppl 1):e000858. doi: 10.1136/bmjgh-2018-000858

61. Ogilvie D, Fayter D, Petticrew M, et al. The harvest plot: a method for synthesising evidence about the differential effects of interventions. *BMC Med Res Methodol* 2008;8:1–8.

62. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;315(7109):629–34.

63. Schünemann HB, Jan; Guyatt, Gordon; Oxman, Andrew. GRADE Handbook: Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach2013.

64. Guyatt GH, Thorlund K, Oxman AD, et al. GRADE guidelines: 13. Preparing summary of findings tables and evidence profiles-continuous outcomes. *J Clin Epidemiol* 2013;66(2):173–83. doi: 10.1016/j.jclinepi.2012.08.001

65. Cochrane. Checklist and Guidance: For disseminating findings from Cochrane intervention reviews: Version 1.0. Project in collaboration with Cochrane Knowledge Translation and Cochrane Norway, 2019.

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Figure legend

Figure 1 Illustration of steps in Qualitative Comparative Analysis

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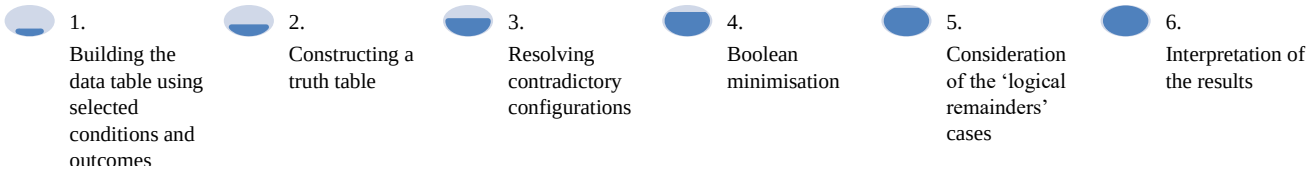


Figure 1 Illustration of steps in Qualitative Comparative Analysis

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	16
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Na
Support:			
Sources	5a	Indicate sources of financial or other support for the review	16
Sponsor	5b	Provide name for the review funder and/or sponsor	16
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	16
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4ff.
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7ff.
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	9f.
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	10f.
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	10f.
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	11
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	11
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	11
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	11
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	11ff.
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	11ff.
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	11ff.
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	11ff.
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	14
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	14

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Keywords:	Dementia < NEUROLOGY, Information technology < BIOTECHNOLOGY & BIOINFORMATICS, GERIATRIC MEDICINE

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Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis

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ABSTRACT

Introduction

Information technology can enhance timely and individual support for people with dementia and informal carers. There is some evidence that people with dementia and informal carers would benefit from technology-based counselling. However, it remains unclear which features of those interventions relate to beneficial outcomes and which aspects are necessary for a successful implementation. Therefore, the primary objectives are: (i) to identify conditions of successful implementation of technology-based counselling interventions in dementia and (ii) to investigate the effectiveness of those interventions.

Methods and analysis

We will conduct a mixed-methods systematic review. The first primary objective requires evidence from various study designs addressing aspects on effective and non-effective implementation of technology-based counselling. This could be telephone-, web- or mobile-based interventions for people with dementia and informal carers. For the second primary objective, randomised controlled trials (RCTs) dealing with any outcomes will be included. Publication year and language will not be restricted. We will search CINAHL, Cochrane Library, MEDLINE, PsycINFO, and Web of Science by April 2021. Additionally, we will perform web searching and citation tracking. To achieve the first primary objective, a Qualitative Comparative Analysis (QCA) will be conducted. The QCA enables us to identify necessary or sufficient components for a successful implementation. To reach the second primary objective, a meta-analysis will be performed with respect to potential clinical and statistical heterogeneity of RCTs. The revised Risk of Bias tool RoB 2.0 will be used to check the risk of bias in RCTs. For all other study designs, the Mixed Method Appraisal Tool will be used.

Ethics and dissemination

Ethics approval is not required for this review. We will disseminate our findings through scientific and non-scientific journal articles and conference presentations as well as formats directed to the public and decision makers in health care.

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Registration details

PROSPERO CRD42021245473

Strengths and limitations of this study

- A methodological strength of our review is the combined determination of effectiveness of technology-based counselling interventions and of successful implementation features of such interventions.
- A comprehensive literature search combining electronic key database search and supplementary search methods with no restrictions on publication date and outcomes is a methodological strength of our review.
- Qualitative Comparative Analysis might uncover effective and non-effective combination(s) of intervention components that might be seen as a methodological strength of our review.
- Stakeholder participation and involvement in the Qualitative Comparative Analysis , the determination of the outcomes of interest for the GRADE approach as well as the interpretation of the review findings enables us to adopt the end-user perspectives and to strengthen the external validity of the review results.
- Probably few studies of technology-based counselling, especially computer-based counselling, could result in limited data for the Qualitative Comparative Analysis.

Keywords

Dementia, Technology-based counselling, Counselling, Implementation, Qualitative Comparative Analysis

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INTRODUCTION

Description of the condition

Dementia is one of the major causes of disability and dependency among older people and is characterized by a progressive decline of cognitive functioning alongside other impairments, such as memory, executive functions, attention, emotional control, and language.[1] Nearly 50 million people have dementia worldwide, of whom approximately 60% live in low- and middle-income countries.[2] Over 9.9 million new cases of dementia are estimated each year worldwide.[3] The average survival time after diagnosis seems to depend on the type of dementia, however, the length of survival is not related to age at diagnosis.[4] Results of a meta-analysis demonstrate an average survival time of 4.11 years after the diagnosis of Lewy body dementia (SD±4.10). People with a diagnosis of Alzheimer's disease have an average survival time of 5.66 years (SD±5.32).[5]

Common consequences of dementia are unmet needs, which can cause behavioural and psychological symptoms.[6] Hyperactivity and apathy show high persistence and incidence, depression and anxiety show low or moderate persistence and moderate incidence, and psychotic symptoms show low persistence and a moderate or low incidence. Around half of all people with dementia need assistance with personal care. According to an estimation by Prince et al. informal carers spent an average of 14 hours weekly to support the person with dementia in activities of daily living (ADLs) such as clothing or bathing. Additionally, another 43 hours for instrumental activities of daily living (IADLs), for instance organisation of care or supervision are needed. No significant differences between countries were identified.[7]

The majority of people with dementia live at home and receive care by a relative, mainly a spouse.[8] Informal carers of people with dementia seem to be less healthy and might experience more difficulties and stress than informal carers of people with other chronic conditions.[9 10] Three dimensions of burden for informal carers have been described: direct impact of caregiving upon informal carers' life (e.g., lost control of life or health suffered), guilt (e.g., should do more or could do a better job), and frustration or embarrassment (e.g., care recipient asks for too much help or carer is embarrassed or angered by the care recipient).[11] Nevertheless, informal caregiving does not only hold a negative connotation, because informal carers feel vindicated in their role or during the development of coping strategies.[12] Excessive informal carers' burden increases the risk of people with dementia having to move from their own home to a nursing home.[13] The

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worldwide COVID-19 (SARS-CoV-2) pandemic raised great concerns for people with dementia related to a lack of information, difficulties in safeguarding procedures or social isolation.[14] Supporting both people with dementia and their informal carers, is essential in order to overcome anxiety, stress, agitation or withdrawal.[15]

Description of the intervention

Information technologies provide supportive options without the risk of infection. Beyond the current pandemic, the intensity of dementia care and the lack of available support services or existing barriers to get access – especially in rural areas – require additional and alternative support possibilities.[16] Information technology can provide timely and individual support for people with dementia and informal carers.[17-23] Various types of support are needed, such as information, education or counselling.[24] Counselling is defined as a talking therapy that involves a trained therapist listening to individuals and helping them to find ways to cope with an emotional issue. It is differentiated from information, education, instruction and case management.[25] Few studies have investigated the effects of counselling for people with dementia and their informal carers.[26] Potential effects include changes in the attitudes of couples confronted with Alzheimer's disease towards becoming more acceptant of the diagnosis along with a more optimistic view for the future.[27]

How the intervention might work

Counselling can be provided through a face-to-face interaction, a group or technological channels, namely: the phone, e-mail, or live chat services. People who received telephone counselling believed that the intervention was attractive with regard to convenience, accessibility, control or inhibition.[28] A recent mixed-methods study uncovered the needs and preferences of informal carers of people with dementia and shows that an online intervention might be an adequate format to provide individual support. From their perspective, tailored programs that are simple to use and interaction with a professional within the online intervention are highly needed.[29] A recent scoping review on virtual support groups for informal carers of people with dementia also highlights the need for tailored, easy-accessible, and moderated online support.[30] Therefore, online counselling might meet the needs of many clients and results in positive outcomes.[31]

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Online counselling can be differentiated from other online services (e.g., web-based education or therapeutic interventions, therapeutic software or online activities, which are not stand-alone services).[32]

Alternatives to face-to-face interventions are crucial in the current COVID-19 pandemic and furthermore have the potential for a more timely and individualised support in dementia. Few systematic reviews investigated technology-based interventions for people with dementia and informal carers (e.g., telephone-counselling,[33] or computer and telephone delivered support interventions [34]). The review by Lins et al. included nine randomised controlled trials (RCTs) and two qualitative studies.[33] The mixed-method review aimed to identify the efficacy of telephone counselling for informal carers of people with dementia and asked for experiences of informal carers with the intervention. The results demonstrate moderate evidence that telephone counselling without other intervention components can reduce depressive symptoms of informal carers. The experiences of informal carers included barriers and facilitators for successful implementation, counsellor's emotional attitude, and content of the telephone counselling. The authors concluded a lack of knowledge related to the aspects of telephone counselling influenced the efficacy of the intervention. The review by Waller et al. included studies related to interventions delivered by a computer (n=10), multiple modalities (n=9), or telephone (n=15).[34] Due to the expanding number of related publications, one could assume an increasing demand to better support informal carers of people with dementia. In conclusion, a need for support and advice for the person with dementia and the informal carer as a dyad was identified.

An overview of systematic reviews investigated psychosocial interventions in dementia and identified counselling and psychotherapy as the second largest category of interventions for informal carers with some evidence to confirm the effectiveness of these interventions.[35] Technology-based interventions were also effective, but components or aims of these interventions were not described. As well as in the reviews by Lins et al. and Waller et al., conditions of telephone-based counselling or technology-based interventions were not described in detail.[33 34]

An update and extension of the review by Lins et al. is timely and warranted.[33] In addition to telephone-based counselling, we will cover other technologies such as computer or web-based interventions. Our review itself will address both the target group of informal carers of people with dementia and studies that

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report on interventions for people with dementia themselves. That step would fulfil the global vision of evidence-based care in dementia, and to providing care and support for the need of people with dementia and their informal carers to enhance their dignity, respect, autonomy and equality.[36]

Objectives

Although there is some evidence that people with dementia and informal carers might benefit from technology-based counselling [33 34], it remains unclear what features of those interventions relate to successful outcomes and what aspects of implementation are necessary and successful. Our review will identify conditions of a successful implementation of technology-based counselling interventions in dementia and investigate the effectiveness of those interventions. The results will present the effectiveness, i.e., beneficial impact and potential adverse events of technology-based counselling for people with dementia and their carers. This is necessary to guide clinical practice concerning the use of such interventions. A comprehensive overview of successful and non-successful intervention components will facilitate a structured guidance for clinical implementation. The objectives of our review are therefore (i) to identify conditions that are aligned with successful implementation of technology-based counselling interventions in dementia and (ii) to investigate the effectiveness of these interventions.

METHODS AND ANALYSIS

We will conduct a mixed-methods systematic review using (i) the methodological approach of Qualitative Comparative Analysis (QCA), aimed at identifying the features of or conditions for successful interventions,[37] and (ii) a meta-analysis, aimed at determining the overall effectiveness of technology-based counselling in dementia. PRISMA-P was used to structure this systematic review protocol.[38]

Eligibility criteria

To reach our review objectives, we are interested in all information concerning the development, evaluation, and implementation of technology-based counselling interventions in dementia. Therefore, we defined the following eligibility criteria.

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Populations

We will include studies on interventions for people with dementia and/or their informal carers . There will be no restrictions concerning setting, gender, age, ethnic background, type and severity of dementia, and relationship between people with dementia and their informal carers. Studies on people with mild cognitive impairment or their informal carers will be excluded. Studies with a mixed sample of informal carers of people with dementia and other conditions will be included if the results for the dementia subgroup have been reported separately. We will also exclude studies on interventions for health and social care professionals.

Interventions

We will include tailored technology-based counselling interventions for people with dementia and/or their informal carers. Technology-based interventions use technologies such as telephone, computer, and web- or mobile-based applications to enable the delivery of the intervention instantly or remotely. Counselling is a complex intervention that might consist of different intervention components or be part of multi-component interventions. To assess the impact of counselling, we will focus on interventions with a core component of counselling. Thus, we will include tailored counselling interventions provided by health and social care professionals such as nurses, physicians, or psychologists aiming to support people with dementia and/or their informal carers (e.g., focusing management of daily living, disease and care management as well as functioning and participation). Therefore, the interventions of interest will be technology-based, provided human-to-human, but may be asynchronous. Studies with interventions that might use counselling as a component of complex care programs and mainly for coordination of care such as case management will be excluded. We will also exclude specific and standardised interventions such as cognitive behaviour therapy, genetic counselling as well as counselling with regard to predictive diagnostics, screening for dementia or diagnostic procedures. Counselling interventions provided by non-professionals (i.e. non-paid) will be excluded.

Comparators

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Technology-based counselling will be compared to no treatment, to standard or usual care, and to in presence or face-to-face counselling. Comparisons between various technology-based interventions will also be considered for inclusion.

Outcomes

We will include all features of effective and non-effective intervention implementation and there will be no restrictions regarding outcomes on the effectiveness of interventions. Our primary outcome is defined as the impact of technology-based counselling on depressive symptoms experienced by people with dementia and their carers. As secondary outcomes, we will consider outcomes on the (i) patient and carer levels (such as burden, well-being, delay or avoidance of transition to a nursing home, decision-making, and acceptance of as well as satisfaction with technology-based counselling) and on the (ii) healthcare system level (such as usage and costs of technology-based counselling).[34 35] We do not expect a negative impact from counselling interventions for people with dementia and carers. However, adverse events will be included.

Design of primary studies

The first primary objective requires evidence from various study designs addressing aspects of implementation success such as qualitative studies, descriptive quantitative studies (e.g., before-after studies, clinical controlled studies, cross-sectional studies, surveys), and mixed-methods research.

To meet the second primary objective, we will include individual and cluster parallel-group RCTs, also RCTs using cross-over and stepped-wedge designs.

Information sources

We will perform a comprehensive literature search combining electronic database search with supplementary search methods with no limitations regarding language and year of publication.[39 40]

Electronic database search

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We will search CINAHL, Cochrane Library including CENTRAL (Cochrane Central Register of Controlled Trials), MEDLINE via PubMed, PsycINFO via Ovid, and the Web of Science Core Collection by April 2021 with no use of filters and limiters.

Supplementary search methods

Google Scholar will be used for free web searching with highly relevant free text terms to identify further studies and conference abstracts not yet identified as well as grey literature to minimise publication bias. We will search further primary studies using (i) hand searching of revealed reviews and (ii) forward and backward citation tracking of included studies using Scopus. To identify ongoing studies, we will search the International Clinical Trials Registry Platform (ICTRP) via CENTRAL, a meta study registry provided by the World Health Organization covering national registries from 17 countries worldwide.

Search update

An update for potentially relevant studies will be conducted in electronic databases one year after the initial search. For the search update, we will use the same search methods and strategies.

Search strategy

We will use database-specific search strategies based on database-specific controlled vocabulary such as MeSH (MEDLINE) and unspecified free text terms. The search strategies will be developed and peer reviewed within the review group using PRESS (Peer Review of Electronic Search Strategies). A researcher experienced in designing search strategies will support us during that process.[41] Our search strategy will contain three components (Dementia, Technology, and Counselling). Search terms and their synonyms will be identified through a previous work by our review group,[33] an orienting search using MEDLINE via PubMed, and the experience of the review group members. Additionally, text mining of titles and abstracts of previously known studies will be applied to identify relevant text terms used in the core literature. Based on the identified search terms, we will look for further search terms using a thesaurus [42] and checking entry terms given in the MeSH browser. For unspecified free text terms, we will search title, abstract, and keywords fields, if available. Controlled vocabulary will be used in the corresponding fields and

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additionally in the title, abstract, and keywords search fields in order to cover publications not yet tagged with controlled vocabulary.[43] The final search strategy per database is provided in the supplementary file.

Study records

Two reviewers will independently screen the identified titles, abstracts, and, if eligible, full texts for inclusion and exclusion using the Rayyan web app.[44] We will discuss conflicting results within the screening team and with a third member of the review group - if necessary. Two reviewers will develop and pilot a data extraction sheet using ten percent of the included studies. Subsequently, one reviewer will extract data. A second reviewer will check data extraction for accuracy. We will discuss conflicting results within the data extracting team and, if necessary, also with a third member of the review group.

Data items

We will extract data on study characteristics (e.g., study design, country), participants' characteristics (e.g., age, gender, type and severity of dementia), intervention and control group characteristics (e.g., type, format, length, frequency), outcome(s) or phenomena of interest (e.g., implantation facilitators and barriers), measurement(s), and results. In addition, we will extract data according to the criteria of TIDieR and CReDECI-2 to provide a description of the reported intervention characteristics and their theoretical background.[45 46]

Outcomes and prioritization

We will involve our advisory board in determining the outcomes of interest following the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach (for details see below).[47]

Risk of bias and critical appraisal in individual studies

We will assess the risk of bias in RCTs using the revised risk of bias tool (RoB 2.0).[48] For all other study designs, we will use the Mixed Method Appraisal Tool, a comprehensive critical appraisal tool provided

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by Hong et al.[49] All appraisals will be conducted by two independent reviewers. We will discuss conflicting results within the appraisal team and, if necessary, with a third member of the review group.

Data synthesis

Qualitative Comparative Analysis (QCA)

To meet our first primary objective, we will conduct a QCA to identify conditions of interventions that are related to successful implementation and effective outcomes. Key elements of the QCA are the configurations of the conditions. These are different scenarios of intervention components and potential facilitators as well as barriers. The QCA enables us to identify necessary or sufficient components for implementation aspects and successful interventions. QCA contains the steps which are displayed in the following sub-chapters.[37]

Building the data table using selected conditions and outcomes

Unlike systematic reviews on the effectiveness interventions, the QCA approach considers various study designs. Studies ("cases" in the QCA terminology) will be added or removed in an iterative process until a later stage of the research.

The selection and definition of the conditions and outcomes is based on the preliminary findings from the literature.[50] The conditions will be grouped into different domains such as setting and participants' characteristics, intervention characteristics, or planned and unplanned measures to facilitate implementation. The number of conditions is expected to be relatively low, and for each condition a clear hypothesis will be formulated according to its expected relation to the outcome.[37] Identified conditions and outcomes will be mapped in a raw data table and calibrated towards their so-called set membership. To prepare the calibration process, a coding framework is needed to provide instructions and values for the data extraction. For each case (= included study), the conditions and outcomes will be determined in crisp-sets with dichotomous values of 0 or 1, or with fuzzy-sets for degrees of membership, transformed in values between 0 and 1.

Constructing a truth table and resolving contradictory configurations

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In a next step, the possible combinations of conditions will be tabulated. Each line in the table represents a configuration of possible conditions using logical operators. The number of configurations will be 2*k, where "k" represents the number of conditions. The configuration represents a potential set where studies can be members or non-members. Studies showing the same configuration will be included in a set and studies showing different configurations will be included in other sets. Therefore, so-called truth tables using the dichotomous or transformed values of crisp- and fuzzy-sets will be developed when configurations (i) are supported by (one or more) cases, (ii) are counter-intuitive or showed an identical combination of conditions, (iii) occurred infrequently, and (iv) triggered positive and negative outcomes at the same time.

Boolean minimisation and consideration of the 'logical remainders' cases

We will subsequently use Boolean minimisation to identify configurations that are observed to adequately lead to a sufficient and specific outcome. We will then consider the plausibility of the configurations and search for consistent conclusions in individual studies (consideration of logical remainders cases). Identified sufficiently specific configurations can form different paths towards an outcome.[50] Therefore, coefficients of coverage will be expressed, and the theoretical relevance will be considered. Furthermore, a "return to the cases" step will help us to evaluate how these results are represented in a study and across studies.[47 51]

Interpretation of the results

The interpreted QCA results will be used for the development of a logical model that can support the practice of technology-based counselling. Furthermore, the newly introduced logical model has the potential to sufficiently inform further research activities in order to compromise inadequate knowledge about the effective steps of implementation. We will use an appropriate software for that analytic step.

Meta-Analysis

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To meet our second primary objective, we will perform a meta-analysis with respect to the potential clinical and statistical heterogeneity of RCTs. Clinical heterogeneity will be indicated by the comparability of interventions based on intervention characteristics and components.[52] Groupings consisting of studies with similar characteristics and interventions might reach comparability. Therefore, it will be possible to conduct stratified meta-analyses given a sufficient number of studies across study groups. We will evaluate the statistical heterogeneity using the I^2 method.[53] A value higher than 50% will be regarded as significantly heterogeneous. Here, we will use the random-effects model and in the case of non-significant heterogeneity, we will use the fixed-effects model to calculate the pooled effect sizes.[54] A p value less than 0.05 will be considered as statistically significant. For meta-analysis of dichotomous results, we will calculate the relative risks (RR) with CI 95%. For the meta-analysis of continuous results, we will use the mean difference (MD) and CI 95%. If different outcome measurements were used, we will use the standardized mean difference (SMD) and CI 95%. To assess the influence of different conditions on the effects of technology-based counselling (e.g., setting, recipient, facilitator, and type of the intervention), we will conduct sensitivity analyses based on subgroup meta-analyses.[55] We will use the software "R" in its current version for our calculations. The results of the meta-analysis will be presented in forest plots.[56]

If the previously described assessments indicate heterogeneous interventions and study results, we will provide a narrative synthesis based on tabular data extraction. Additionally, we will summarise effects, risk of bias, intervention and study characteristics graphically by means of a Harvest plot to report appropriately on the complexity and heterogeneity of the included studies.[57 58]

Meta-bias

Given a sufficient number of included studies to meet our second primary objective, risk for publication bias will be assessed graphically using a funnel plot complemented by a statistical test.[59]

Confidence in cumulative evidence

The quality of evidence will be assessed using the GRADE approach by rating the quality of evidence as high, moderate, low or very low.[60] The combined results of meta-analyses and the GRADE approach will be presented in a summary of findings table.[61]

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Patient and Public Involvement

We will include representatives of people with dementia and informal carers, i.e. a person engaged in a support organisation that supports dementia care, and an informal carer of a person with dementia. The advisory board will consist of two or three people. Recommendations by the advisory board will be used for the qualitative steps of the QCA, where experience and knowledge of the subject are needed for the identification and the assessment of conditions that are important for the intervention. Furthermore, the advisory board will be involved in determining the outcomes of interest for the GRADE approach as well as the interpretation of the review findings.[47]

ETHICS AND DISSEMINATION

Ethics approval is not required for this review protocol. The members of the advisory board participate voluntarily in the review process. Ongoing informed consent will be obtained.

Our dissemination strategy uses multiple channels of distribution to share our review work and results with academic and non-academic stakeholders, the society, and interested individuals.[62]

The final review will be published in an international open access journal relevant in the field of dementia care. Additionally, we will discuss our review results with experts at national and international conferences. Plain language summaries of the review will be placed on our institutional websites and social media profiles (e.g., Twitter, ResearchGate). A second publication will be published in a non-academic journal to share our review results with other interested groups such as practitioners and carers of people with dementia. Furthermore, we will participate in interactive mass media platforms to enable groups of interest to ask questions. A cartoon video will be accessible and disseminated online. Additionally, the results will be printed in newsletters of organisations representing people with dementia as well as patient advocacy groups such as Deutsche Alzheimer Gesellschaft, Alzheimer Schweiz, Österreichische Alzheimer Gesellschaft, and Alzheimer Europe. Furthermore, we will contact insurance companies and other stakeholders to disseminate our research results.

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Authors' contributions

Substantial contributions to the conception or design of the work: JH, GL, GM, AB

Drafting the work: JH, AB

Revising it critically for important intellectual content: JH, GL, FW, DB, GM, AB

Final approval of the version to be published: JH, GL, FW, DB, GM, AB

Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy and integrity of any part of the work are appropriately investigated and resolved: JH, GL, FW, DB, GM, AB

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Competing interests' statement

All authors declare that there is no competing interest.

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REFERENCES

1. WHO. ICD-11 for Mortality and Morbidity Statistics. Neurocognitive disorders 2019 [Available from: https://www.who.int/classifications/icd/en/.
2. WHO. Dementia, 2019.
3. Prince M, Wimo A, Guerchet M, et al. World Alzheimer Report 2015: The Global Impact of Dementia: An Analysis of Prevalence, Incidence, Cost and Trend London2015 [Available from: https://www.alz.co.uk/research/WorldAlzheimerReport2015.pdf accessed 19.03.2020.
4. Brodaty H, Seeher K, Gibson L. Dementia time to death: a systematic literature review on survival time and years of life lost in people with dementia. *Int Psychogeriatr* 2012;24(7):1034-45. doi: 10.1017/S1041610211002924 [published Online First: 2012/02/14]
5. Mueller C, Soysal P, Rongve A, et al. Survival time and differences between dementia with Lewy bodies and Alzheimer's disease following diagnosis: A meta-analysis of longitudinal studies. *Ageing Res Rev* 2019;50:72-80. doi: 10.1016/j.arr.2019.01.005 [published Online First: 2019/01/10]
6. Hung L, Lee PA, Au-Yeung AT, et al. Adopting a Clinical Assessment Framework in Older Adult Mental Health. *J Psychosoc Nurs Ment Health Serv* 2016;54(7):26–31. doi: 10.3928/02793695-20160616-05
7. Prince M, Prina M, Guerchet M. World Alzheimer Report 2013: Journey of Caring: An analysis of long-term care for dementia London2013 [Available from: https://www.alz.co.uk/research/WorldAlzheimerReport2013.pdf accessed 19.03.2020.
8. Wimo A, Gauthier S, Prince M. Global estimates of informal care 2018 [Available from: https://www.alz.co.uk/adi/pdf/global-estimates-of-informal-care.pdf accessed 15.05.2020.
9. Schoenmakers B, Buntinx F, Delepeleire J. Factors determining the impact of care-giving on caregivers of elderly patients with dementia. A systematic literature review. *Maturitas* 2010;66(2):191-200. doi: 10.1016/j.maturitas.2010.02.009 [published Online First: 2010/03/24]
10. Isik AT, Soysal P, Solmi M, et al. Bidirectional relationship between caregiver burden and neuropsychiatric symptoms in patients with Alzheimer's disease: A narrative review. *Int J Geriatr Psychiatry* 2019;34(9):1326-34. doi: 10.1002/gps.4965 [published Online First: 2018/09/11]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

11. Springate BA, Tremont G. Dimensions of caregiver burden in dementia: impact of demographic, mood, and care recipient variables. *Am J Geriatr Psychiatry* 2014;22(3):294-300. doi: 10.1016/j.jagp.2012.09.006 [published Online First: 2013/04/10]

12. Lloyd J, Patterson T, Muers J. The positive aspects of caregiving in dementia: A critical review of the qualitative literature. *Dementia* 2016;15(6):1534-61. doi: 10.1177/1471301214564792 [published Online First: 2016/10/30]

13. Cepoiu-Martin M, Tam-Tham H, Patten S, et al. Predictors of long-term care placement in persons with dementia: a systematic review and meta-analysis. *Int J Geriatr Psychiatry* 2016;31(11):1151-71. doi: 10.1002/gps.4449 [published Online First: 2016/10/19]

14. Wang H, Li T, Barbarino P, et al. Dementia care during COVID-19. *The Lancet* 2020;395(10231):1190-91. doi: 10.1016/s0140-6736(20)30755-8

15. WHO. ADI offers advice and support during COVID-19 https://www.alz.co.uk/news/adi-offers-advice-and-support-during-covid-192020 [Available from: https://www.alz.co.uk/news/adi-offers-advice-and-support-during-covid-19 accessed 05.05.2020.

16. Innes A, Morgan D, Kosteniuk J. Dementia care in rural and remote settings: a systematic review of informal/family caregiving. *Maturitas* 2011;68(1):34-46. doi: 10.1016/j.maturitas.2010.10.002 [published Online First: 2010/11/26]

17. Bejan A, Gündogdu R, Butz K, et al. Using multimedia information and communication technology (ICT) to provide added value to reminiscence therapy for people with dementia: Lessons learned from three field studies. *Z Gerontol Geriatr* 2017;51(1):9–15. doi: 10.1007/s00391-017-1347-7

18. D'Onofrio G, Sancarlo D, Ricciardi F, et al. Information and Communication Technologies for the Activities of Daily Living in Older Patients with Dementia: A Systematic Review. *J Alzheimer's Dis* 2017;Article in Press doi: 10.3233/JAD-161145

19. Lauriks S, Reinersmann A, van der Roest HG, et al. Review of ICT-based services for identified unmet needs in people with dementia. *Ageing Res Rev* 2007;6(3):223–46. doi: 10.1016/j.arr.2007.07.002

20. Lucero RJ, Fehlberg EA, Patel AGM, et al. The effects of information and communication technologies on informal caregivers of persons living with dementia: A systematic review. *Alzheimers Dement* 2019;5:1–12. doi: 10.1016/j.trci.2018.11.003

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

21. McHugh JE, Wherton JP, Prendergast DK, et al. Identifying opportunities for supporting caregivers of persons with dementia through information and communication technology. *Gerontechnology* 2013;10(4) doi: 10.4017/gt.2012.10.4.003.00

22. Noone C, McSharry J, Smalle M, et al. Video calls for reducing social isolation and loneliness in older people: a rapid review. *Cochrane Database Syst Rev* 2020;5:CD013632. doi: 10.1002/14651858.CD013632

23. Pleasant M, Molinari V, Dobbs D, et al. Effectiveness of online dementia caregivers training programs: A systematic review. *Geriatr Nurs* 2020;Article in Press doi: 10.1016/j.gerinurse.2020.07.004

24. Whitlatch CJ, Orsulic-Jeras S. Meeting the Informational, Educational, and Psychosocial Support Needs of Persons Living With Dementia and Their Family Caregivers. *Gerontologist* 2018;58(suppl_1):S58-S73. doi: 10.1093/geront/gnx162 [published Online First: 2018/01/24]

25. NHS. Counselling https://www.nhs.uk/conditions/counselling/2017 [accessed 17.1.2020.

26. Dickinson C, Dow J, Gibson G, et al. Psychosocial intervention for carers of people with dementia: What components are most effective and when? A systematic review of systematic reviews. *Int Psychogeriatr* 2017;29(1):31-43. doi: 10.1017/S1041610216001447 [published Online First: 2016/09/27]

27. Elvish R, Lever S-J, Johnstone J, et al. Psychological interventions for carers of people with dementia: A systematic review of quantitative and qualitative evidence. *Counsell Psychother Res J* 2013;13(2):106-25. doi: 10.1080/14733145.2012.739632

28. Reese RJ, Conoley C, W., Brossart DF. The Attractiveness of Telephone Counseling: An Empirical Investigation of Client Perceptions. *J Couns Dev* 2006;84:54-60.

29. Teles S, Paúl C, Sosa Napolskij M, et al. Dementia caregivers training needs and preferences for online interventions: a mixed-methods study. *J Clin Nurs* 2020;Article in Press doi: 10.1111/jocn.15445

30. Armstrong MJ, Alliance S. Virtual Support Groups for Informal Caregivers of Individuals With Dementia: A Scoping Review. *Alzheimer Dis Assoc Disord* 2019;33(4):362–69. doi: 10.1097/WAD.0000000000000349

31. Harris B, Birnbaum R. Ethical and Legal Implications on the Use of Technology in Counselling. *Clin Soc Work J* 2015;43:133–41.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

32. Dowling M, Rickwood D. Online Counseling and Therapy for Mental Health Problems: A Systematic Review of Individual Synchronous Interventions Using Chat. *J Tech Hum Serv* 2013;31(1):1-21.

33. Lins S, Hayder-Beichel D, Ruecker G, et al. Efficacy and experiences of telephone counselling for informal carers of people with dementia. *Cochrane Database Syst Rev* 2014(9):CD009126. doi: 10.1002/14651858.CD009126.pub2

34. Waller A, Dilworth S, Mansfield E, et al. Computer and telephone delivered interventions to support caregivers of people with dementia: a systematic review of research output and quality. *BMC Geriatr* 2017;17(1):265. doi: 10.1186/s12877-017-0654-6 [published Online First: 2017/11/18]

35. Clarkson P, Hughes J, Xie C, et al. Overview of systematic reviews: Effective home support in dementia care, components and impacts-Stage 1, psychosocial interventions for dementia. *J Adv Nurs* 2017;73(12):2845-63. doi: 10.1111/jan.13362

36. WHO. Global action plan on the public health response to dementia 2017 - 2025. 2017

37. Rihoux B, Ragin CC, editors. *Configurational comparative methods: Qualitative comparative analysis (QCA) and related techniques*. Los Angeles: SAGE, 2009.

38. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1. doi: 10.1186/2046-4053-4-1

39. Cooper C, Booth A, Varley-Campbell J, et al. Defining the process to literature searching in systematic reviews: a literature review of guidance and supporting studies. *BMC Med Res Methodol* 2018;18:85. doi: 10.1186/s12874-018-0545-3

40. Lefebvre C, Glanville J, Briscoe S, et al. Searching for and selecting studies. In: Higgins JPT, Thomas J, eds. Cochrane Handbook for Systematic Reviews of Interventions. 2 ed. Hoboken: Wiley Online Library 2019:67–108.

41. McGowan J, Sampson M, Salzwedel DM, et al. PRESS Peer Review of Electronic Search Strategies: 2015 Guideline Statement. *J Clin Epidemiol* 2016;75:40–46.

42. Dictionary.com. Thesaurus.com 2020 [Available from: http://www.thesaurus.com/ accessed 28.04.2020.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

43. Rodriguez RW. Comparison of indexing times among articles from medical, nursing, and pharmacy journals. *Am J Health Syst Pharm* 2016;73(8):569–75. doi: 10.2146/ajhp150319

44. Ouzzani M, Hammady H, Fedorowicz Z, et al. Rayyan-a web and mobile app for systematic reviews. *Syst Rev* 2016;5:210. doi: 10.1186/s13643-016-0384-4

45. Hoffmann TC, Glasziou P, Boutron I, et al. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *BMJ* 2014;348:g1687.

46. Möhler R, Köpke S, Meyer G. Criteria for Reporting the Development and Evaluation of Complex Interventions in healthcare: revised guideline (CReDECI 2). *Trials* 2015;16(1):204.

47. Langer G, Meerpohl JJ, Perleth M, et al. GRADE-Leitlinien: 2. Formulierung der Fragestellung und Entscheidung über wichtige Endpunkte. *Z Evid Fortbild Qual Gesundhwes* 2012;106(5):369–76.

48. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;2:l4898. doi: 10.1136/bmj.l4898

49. Hong QN, Fàbregues S, Bartlett G, et al. The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. *Education for Information* 2018;34(4):285–91. doi: 10.3233/EFI-180221

50. Schneider CQ, Wagemann C. Standards of Good Practice in Qualitative Comparative Analysis (QCA) and Fuzzy-Sets. *Comp Sociol* 2010;9(3):397–418. doi: 10.1163/156913210X12493538729793

51. Kahwati L, Jacobs S, Kane H, et al. Using qualitative comparative analysis in a systematic review of a complex intervention. *Syst Rev* 2016;5:82. doi: 10.1186/s13643-016-0256-y

52. West SL, Gartlehner G, Mansfield AJ, et al. Comparative Effectiveness Review Methods: Clinical Heterogeneity. Agency for Healthcare Research and Quality. Rockville, 2010.

53. Higgins JPT, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557–60. doi: 10.1136/bmj.327.7414.557

54. Hedges LV, Vevea J. Fixed-and random-effects models in meta-analysis. *Psychol Methods* 1998;3(4):486–504.

55. Khan KS, Kunz R, Kleijnen J, et al. Systematic Reviews to Support Evidence Bsaed Medicine. How to Review and Apply Findings of Healthcare Research London: Hodder & Stoughton.

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56. Lewis S, Clarke M. Forest plots: trying to see the wood and the trees. *BMJ* 2001;322(7300):1479–80. doi: 10.1136/bmj.322.7300.1479

57. Higgins JPT, López-López JA, Becker BJ, et al. Synthesising quantitative evidence in systematic reviews of complex health interventions. *BMJ Glob Health* 2019;4(Suppl 1):e000858. doi: 10.1136/bmjgh-2018-000858

58. Ogilvie D, Fayter D, Petticrew M, et al. The harvest plot: a method for synthesising evidence about the differential effects of interventions. *BMC Med Res Methodol* 2008;8:1–8.

59. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;315(7109):629–34.

60. Schünemann HB, Jan; Guyatt, Gordon; Oxman, Andrew. GRADE Handbook: Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach2013.

61. Guyatt GH, Thorlund K, Oxman AD, et al. GRADE guidelines: 13. Preparing summary of findings tables and evidence profiles-continuous outcomes. *J Clin Epidemiol* 2013;66(2):173–83. doi: 10.1016/j.jclinepi.2012.08.001

62. Cochrane. Checklist and Guidance: For disseminating findings from Cochrane intervention reviews: Version 1.0. Project in collaboration with Cochrane Knowledge Translation and Cochrane Norway, 2019.

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Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis

Supplement: search strategies

MEDLINE via PubMed

Date: 20.04.2021

#	Entry	Hits
1	"DEMENT*"[TITLE/ABSTRACT] OR "ALZHEIMER*"[TITLE/ABSTRACT] OR "MAJOR NEUROCOGNITIVE DISORDER*"[TITLE/ABSTRACT] OR "DEMENTIA"[MESH TERMS]	273374
2	"APP"[TITLE/ABSTRACT] OR "APPS"[TITLE/ABSTRACT] OR "AUGMENTED REALIT*"[TITLE/ABSTRACT] OR "BLENDED TREATMENT"[TITLE/ABSTRACT] OR "BLENDED THERAPY"[TITLE/ABSTRACT] OR "BLOG"[TITLE/ABSTRACT] OR "CELL PHONE*"[TITLE/ABSTRACT] OR "CELLPHONE*"[TITLE/ABSTRACT] OR "CHAT"[TITLE/ABSTRACT] OR "CHATS"[TITLE/ABSTRACT] OR "CHATBOT*"[TITLE/ABSTRACT] OR "COMPUTER*"[TITLE/ABSTRACT] OR "DIGITAL*"[TITLE/ABSTRACT] OR "E COUNSELING"[TITLE/ABSTRACT] OR "ECOUNSELING"[TITLE/ABSTRACT] OR "E COUNSELING"[TITLE/ABSTRACT] OR "E COUNSELLING"[TITLE/ABSTRACT] OR "ECOUNSELLING"[TITLE/ABSTRACT] OR "E COUNSELLING"[TITLE/ABSTRACT] OR "E HEALTH"[TITLE/ABSTRACT] OR "EHEALTH"[TITLE/ABSTRACT] OR "E HEALTH"[TITLE/ABSTRACT] OR "ELECTRONIC*"[TITLE/ABSTRACT] OR "EMAIL*"[TITLE/ABSTRACT] OR "E MAIL*"[TITLE/ABSTRACT] OR "FACEBOOK"[TITLE/ABSTRACT] OR "ICT"[TITLE/ABSTRACT] OR "INTERNET"[TITLE/ABSTRACT] OR "MESSENGER"[TITLE/ABSTRACT] OR "MHEALTH"[TITLE/ABSTRACT] OR "M-HEALTH"[TITLE/ABSTRACT] OR "MOBILE DEVICE"[TITLE/ABSTRACT] OR "MOBILE HEALTH"[TITLE/ABSTRACT] OR "MOBILE PHONE"[TITLE/ABSTRACT] OR "MOBILE HEALTH"[TITLE/ABSTRACT] OR "ONLINE*"[TITLE/ABSTRACT] OR "PHONE"[TITLE/ABSTRACT] OR "REMOTE"[TITLE/ABSTRACT] OR "SKYPE"[TITLE/ABSTRACT] OR "SMARTPHONE*"[TITLE/ABSTRACT] OR "SMS"[TITLE/ABSTRACT] OR "SOCIAL MEDIA"[TITLE/ABSTRACT] OR "TABLET"[TITLE/ABSTRACT] OR "TABLETS"[TITLE/ABSTRACT] OR "TECHNOLOG*"[TITLE/ABSTRACT] OR "TELE HOME CARE"[TITLE/ABSTRACT] OR "TELE MEDICINE"[TITLE/ABSTRACT] OR "TELECARE"[TITLE/ABSTRACT] OR "TELE-CARE"[TITLE/ABSTRACT] OR "TELECARING"[TITLE/ABSTRACT] OR "TELE-CARING"[TITLE/ABSTRACT] OR "TELECOMMUNICATION"[TITLE/ABSTRACT] OR "TELECONFERENCE*"[TITLE/ABSTRACT] OR "TELE CONFERENCE*"[TITLE/ABSTRACT] OR "TELECONSULTATION*"[TITLE/ABSTRACT] OR "TELE CONSULTATION*"[TITLE/ABSTRACT] OR "TELEHEALTH"[TITLE/ABSTRACT] OR "TELE-HEALTH"[TITLE/ABSTRACT] OR "TELEMEDICINE"[TITLE/ABSTRACT] OR "TELENURSE"[TITLE/ABSTRACT] OR "TELENURSING"[TITLE/ABSTRACT] OR "TELEPHONE"[TITLE/ABSTRACT] OR "TELEPHONES"[TITLE/ABSTRACT] OR "TELEPRESENCE"[TITLE/ABSTRACT] OR "TELEPRESENT"[TITLE/ABSTRACT] OR "TELEREHABILITATION"[TITLE/ABSTRACT] OR "TELE-REHABILITATION"[TITLE/ABSTRACT] OR "TELETHERAP*"[TITLE/ABSTRACT] OR "TELE THERAP*"[TITLE/ABSTRACT] OR "TELETREATMENT"[TITLE/ABSTRACT] OR "TELE-TREATMENT"[TITLE/ABSTRACT] OR "VIDEOCONFERENCE"[TITLE/ABSTRACT] OR "VIDEOCONFERENCES"[TITLE/ABSTRACT] OR "VIRTUAL"[TITLE/ABSTRACT] OR "WEB BASED"[TITLE/ABSTRACT] OR "WEBBASED"[TITLE/ABSTRACT] OR "WEB BASED"[TITLE/ABSTRACT] OR "WORLD WIDE WEB"[TITLE/ABSTRACT] OR "TELEMEDICINE"[MESH TERMS] OR "TELENURSING"[MESH TERMS] OR "TECHNOLOGY"[MESH TERMS] OR "VIRTUAL REALITY"[MESH TERMS] OR "TELECOMMUNICATIONS"[MESH TERMS] OR "VIDEOCONFERENCING"[MESH TERMS] OR "TELEREHABILITATION"[MESH TERMS] OR "INTERNET-BASED INTERVENTION"[MESH TERMS] OR "SOCIAL MEDIA"[MESH TERMS] OR "INFORMATION TECHNOLOGY"[MESH TERMS] OR "REMOTE CONSULTATION"[MESH TERMS] OR "DISTANCE COUNSELING"[MESH TERMS] OR "VIRTUAL REALITY"[MESH TERMS] OR "TELEPHONE"[MESH TERMS] OR "CELL PHONE"[MESH TERMS]	2123169
3	"ADVICE*"[TITLE/ABSTRACT] OR "ADVISE*"[TITLE/ABSTRACT] OR "ADVISING"[TITLE/ABSTRACT] OR "COACH*"[TITLE/ABSTRACT] OR "CONSULT*"[TITLE/ABSTRACT] OR "COUNSELLING"[TITLE/ABSTRACT] OR "COUNSELING"[TITLE/ABSTRACT] OR "INSTRUCT*"[TITLE/ABSTRACT] OR "GUIDANCE*"[TITLE/ABSTRACT] OR "GUIDE*"[TITLE/ABSTRACT] OR "MENTORING"[TITLE/ABSTRACT] OR "PASTORAL CARE"[TITLE/ABSTRACT] OR "SOCIAL SUPPORT*"[TITLE/ABSTRACT] OR "SUPERVIS*"[TITLE/ABSTRACT] OR "SUPPORT SERVICE*"[TITLE/ABSTRACT] OR "TALKING THERAP*"[TITLE/ABSTRACT] OR "MENTORING"[MESH TERMS] OR "COUNSELING"[MESH TERMS] OR "SOCIAL SUPPORT"[MESH TERMS]	1362940
4	1 AND 2 AND 3	1991
5	(SURGICAL OR SURGERY OR GENETI* OR PROTEOMIC* OR GENOMIC* OR PROTEIN* OR NEUROIMAG*).TI,AB.	6115294
6	4 NOT 5	1623

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#	Entry	Hits
1	TI DEMENT* OR TI ALZHEIMER* OR TI "MAJOR NEUROCOGNITIVE DISORDER*" OR AB DEMENT* OR AB ALZHEIMER* OR AB "MAJOR NEUROCOGNITIVE DISORDER*" OR MH DEMENTIA+	97882
2	TI APP OR TI APPS OR TI "AUGMENTED REALIT*" OR TI "BLENDED TREATMENT" OR TI "BLENDED THERAPY" OR TI BLOG OR TI "CELL PHONE*" OR TI CELLPHONE* OR TI CHAT OR TI CHATS OR TI CHATBOT* OR TI COMPUTER* OR TI DIGITAL* OR TI "E COUNSELING" OR TI ECOUNSELING OR TI E-COUNSELING OR TI "E COUNSELLING" OR TI ECOUNSELLING OR TI E-COUNSELLING OR TI "E HEALTH" OR TI EHEALTH OR TI E-HEALTH OR TI ELECTRONIC* OR TI EMAIL* OR TI E-MAIL* OR TI FACEBOOK OR TI ICT OR TI INTERNET OR TI MESSENGER OR TI MHEALTH OR TI M-HEALTH OR TI "MOBILE DEVICE" OR TI "MOBILE HEALTH" OR TI "MOBILE PHONE" OR TI "MOBILE-HEALTH" OR TI ONLINE* OR TI PHONE OR TI REMOTE OR TI SKYPE OR TI SMARTPHONE* OR TI SMS OR TI "SOCIAL MEDIA" OR TI TABLET OR TI TABLETS OR TI TECHNOLOG* OR TI "TELE HOME CARE" OR TI "TELE MEDICINE" OR TI TELECARE OR TI TELE-CARE OR TI TELECARING OR TI TELE-CARING OR TI TELECOMMUNICATION OR TI TELECONFERENCE* OR TI TELE-CONFERENCE* OR TI TELECONSULTATION* OR TI TELE-CONSULTATION* OR TI TELEHEALTH OR TI TELE-HEALTH OR TI TELEMEDICINE OR TI TELENURSE OR TI TELENURSING OR TI TELEPHONE OR TI TELEPHONES OR TI TELEPRESENCE OR TI TELEPRESENT OR TI TELEREHABILITATION OR TI TELE-REHABILITATION OR TI TELETHERAP* OR TI TELE-THERAP* OR TI TELETREATMENT OR TI TELE-TREATMENT OR TI VIDEOCONFERENCE OR TI VIDEOCONFERENCES OR TI VIRTUAL OR TI "WEB BASED" OR TI WEBBASED OR TI WEB-BASED OR TI "WORLD WIDE WEB" OR AB APP OR AB APPS OR AB "AUGMENTED REALIT*" OR AB "BLENDED TREATMENT" OR AB "BLENDED THERAPY" OR AB BLOG OR AB "CELL PHONE*" OR AB CELLPHONE* OR AB CHAT OR AB CHATS OR AB CHATBOT* OR AB COMPUTER* OR AB DIGITAL* OR AB "E COUNSELING" OR AB ECOUNSELING OR AB E-COUNSELING OR AB "E COUNSELLING" OR AB ECOUNSELLING OR AB E-COUNSELLING OR AB "E HEALTH" OR AB EHEALTH OR AB E-HEALTH OR AB ELECTRONIC* OR AB EMAIL* OR AB E-MAIL* OR AB FACEBOOK OR AB ICT OR AB INTERNET OR AB MESSENGER OR AB MHEALTH OR AB M-HEALTH OR AB "MOBILE DEVICE" OR AB "MOBILE HEALTH" OR AB "MOBILE PHONE" OR AB "MOBILE-HEALTH" OR AB ONLINE* OR AB PHONE OR AB REMOTE OR AB SKYPE OR AB SMARTPHONE* OR AB SMS OR AB "SOCIAL MEDIA" OR AB TABLET OR AB TABLETS OR AB TECHNOLOG* OR AB "TELE HOME CARE" OR AB "TELE MEDICINE" OR AB TELECARE OR AB TELE-CARE OR AB TELECARING OR AB TELE-CARING OR AB TELECOMMUNICATION OR AB TELECONFERENCE* OR AB TELE-CONFERENCE* OR AB TELECONSULTATION* OR AB TELE-CONSULTATION* OR AB TELEHEALTH OR AB TELE-HEALTH OR AB TELEMEDICINE OR AB TELENURSE OR AB TELENURSING OR AB TELEPHONE OR AB TELEPHONES OR AB TELEPRESENCE OR AB TELEPRESENT OR AB TELEREHABILITATION OR AB TELE-REHABILITATION OR AB TELETHERAP* OR AB TELE-THERAP* OR AB TELETREATMENT OR AB TELE-TREATMENT OR AB VIDEOCONFERENCE OR AB VIDEOCONFERENCES OR AB VIRTUAL OR AB "WEB BASED" OR AB WEBBASED OR AB WEB-BASED OR AB "WORLD WIDE WEB" OR MH TELEMEDICINE+ OR MH TELENURSING+ OR MH TECHNOLOGY+ OR MH VIRTUAL REALITY+ OR MH TELECOMMUNICATIONS+ OR MH VIDEOCONFERENCING+ OR MH TELEREHABILITATION+ OR MH INTERNET-BASED INTERVENTION+ OR MH SOCIAL MEDIA+ OR MH INFORMATION TECHNOLOGY+ OR MH REMOTE CONSULTATION+ OR MH TELEPHONE+ OR MH CELL PHONE+	612076
3	TI ADVICE* OR TI ADVISE* OR TI ADVISING OR TI COACH* OR TI CONSULT* OR TI COUNSELLING OR TI COUNSELING OR TI INSTRUCT* OR TI GUIDANCE* OR TI GUIDE* OR TI MENTORING OR TI "PASTORAL CARE" OR TI "SOCIAL SUPPORT*" OR TI SUPERVIS* OR TI "SUPPORT SERVICE*" OR TI "TALKING THERAP*" OR AB ADVICE* OR AB ADVISE* OR AB ADVISING OR AB COACH* OR AB CONSULT* OR AB COUNSELLING OR AB COUNSELING OR AB INSTRUCT* OR AB GUIDANCE* OR AB GUIDE* OR AB MENTORING OR AB "PASTORAL CARE" OR AB "SOCIAL SUPPORT*" OR AB SUPERVIS* OR AB "SUPPORT SERVICE*" OR AB "TALKING THERAP*" OR MH COUNSELING+ OR MH PSYCHOSOCIAL SUPPORT+	575744
4	1 AND 2 AND 3	962
5	TI SURGICAL OR TI SURGERY OR TI GENETI* OR TI PROTEOMIC* OR TI GENOMIC* OR TI PROTEIN* OR TI NEUROIMAG* OR AB SURGICAL OR AB SURGERY OR AB GENETI* OR AB PROTEOMIC* OR AB GENOMIC* OR AB PROTEIN* OR AB NEUROIMAG*	643710
6	4 NOT 5	895

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1	(DEMENT* OR ALZHEIMER* OR "MAJOR NEUROCOGNITIVE DISORDER*"):TI,AB,KW	20011
2	EINGABE: SUCHKOMP (APP OR APPS OR "AUGMENTED REALIT*" OR "BLENDED TREATMENT" OR "BLENDED THERAPY" OR BLOG OR "CELL PHONE*" OR CELLPHONE* OR CHAT OR CHATS OR CHATBOT* OR COMPUTER* OR DIGITAL* OR "E COUNSELING" OR ECOUNSELING OR E-COUNSELING OR "E COUNSELLING" OR ECOUNSELLING OR E-COUNSELLING OR "E HEALTH" OR EHEALTH OR E-HEALTH OR ELECTRONIC* OR EMAIL* OR E-MAIL* OR FACEBOOK OR ICT OR INTERNET OR MESSENGER OR MHEALTH OR M-HEALTH OR "MOBILE DEVICE" OR "MOBILE HEALTH" OR "MOBILE PHONE" OR "MOBILE-HEALTH" OR ONLINE* OR PHONE OR REMOTE OR SKYPE OR SMARTPHONE* OR SMS OR "SOCIAL MEDIA" OR TABLET OR TABLETS OR TECHNOLOG* OR "TELE HOME CARE" OR "TELE MEDICINE" OR TELECARE OR TELE-CARE OR TELECARING OR TELE-CARING OR TELECOMMUNICATION OR TELECONFERENCE* OR TELE-CONFERENCE* OR TELECONSULTATION* OR TELE-CONSULTATION* OR TELEHEALTH OR TELE-HEALTH OR TELEMEDICINE OR TELENURSE OR TELENURSING OR TELEPHONE OR TELEPHONES OR TELEPRESENCE OR TELEPRESENT OR TELEREHABILITATION OR TELE-REHABILITATION OR TELETHERAP* OR TELE-THERAP* OR TELETREATMENT OR TELE-TREATMENT OR VIDEOCONFERENCE OR VIDEOCONFERENCES OR VIRTUAL OR "WEB BASED" OR WEBBASED OR WEB-BASED OR "WORLD WIDE WEB"):TI,AB,KW ONENTE 2	193349
3	(ADVICE* OR ADVISE* OR ADVISING OR COACH* OR CONSULT* OR COUNSELLING OR COUNSELING OR INSTRUCT* OR GUIDANCE* OR GUIDE* OR MENTORING OR "PASTORAL CARE" OR "SOCIAL SUPPORT*" OR SUPERVIS* OR "SUPPORT SERVICE*" OR "TALKING THERAP*"):TI,AB,KW	168646
4	1 AND 2 AND 3	802
5	(SURGICAL OR SURGERY OR GENETI* OR PROTEOMIC* OR GENOMIC* OR PROTEIN* OR NEUROIMAG*):TI,AB,KW	359548
6	4 NOT 5	692

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#	Entry	Hits
1	TS=(DEMENT* OR ALZHEIMER* OR "MAJOR NEUROCOGNITIVE DISORDER*") INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	343520
2	TS=(APP OR APPS OR "AUGMENTED REALIT*" OR "BLENDED TREATMENT" OR "BLENDED THERAPY" OR BLOG OR "CELL PHONE*" OR CELLPHONE* OR CHAT OR CHATS OR CHATBOT* OR COMPUTER* OR DIGITAL* OR "E COUNSELING" OR ECOUNSELING OR E-COUNSELING OR "E COUNSELLING" OR ECOUNSELLING OR E-COUNSELLING OR "E HEALTH" OR EHEALTH OR E-HEALTH OR ELECTRONIC* OR EMAIL* OR E-MAIL* OR FACEBOOK OR ICT OR INTERNET OR MESSENGER OR MHEALTH OR M-HEALTH OR "MOBILE DEVICE" OR "MOBILE HEALTH" OR "MOBILE PHONE" OR "MOBILE-HEALTH" OR ONLINE* OR PHONE OR REMOTE OR SKYPE OR SMARTPHONE* OR SMS OR "SOCIAL MEDIA" OR TABLET OR TABLETS OR TECHNOLOG* OR "TELE HOME CARE" OR "TELE MEDICINE" OR TELECARE OR TELE-CARE OR TELECARING OR TELE-CARING OR TELECOMMUNICATION OR TELECONFERENCE* OR TELE-CONFERENCE* OR TELECONSULTATION* OR TELE-CONSULTATION* OR TELEHEALTH OR TELE-HEALTH OR TELEMEDICINE OR TELENURSE OR TELENURSING OR TELEPHONE OR TELEPHONES OR TELEPRESENCE OR TELEPRESENT OR TELEREHABILITATION OR TELE-REHABILITATION OR TELETHERAP* OR TELE-THERAP* OR TELETREATMENT OR TELE-TREATMENT OR VIDEOCONFERENCE OR VIDEOCONFERENCES OR VIRTUAL OR "WEB BASED" OR WEBBASED OR WEB-BASED OR "WORLD WIDE WEB") INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	5975986
3	TS=(ADVICE* OR ADVISE* OR ADVISING OR COACH* OR CONSULT* OR COUNSELLING OR COUNSELING OR INSTRUCT* OR GUIDANCE* OR GUIDE* OR MENTORING OR "PASTORAL CARE" OR "SOCIAL SUPPORT*" OR SUPERVIS* OR "SUPPORT SERVICE*" OR "TALKING THERAP*") INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	2256839
4	1 AND 2 AND 3	2743
5	TS=(SURGICAL OR SURGERY OR GENETI* OR PROTEOMIC* OR GENOMIC* OR PROTEIN* OR NEUROIMAG*) INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	7195603
6	4 NOT 5	2183

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#	Entry	Hits
1	(DEMENT* OR ALZHEIMER* OR "MAJOR NEUROCOGNITIVE DISORDER*").TI,AB,MP. OR EXP DEMENTIA/ OR EXP ALZHEIMER DISEASE/	113751
2	(APP OR APPS OR "AUGMENTED REALIT*" OR "BLENDED TREATMENT" OR "BLENDED THERAPY" OR BLOG OR "CELL PHONE*" OR CELLPHONE* OR CHAT OR CHATS OR CHATBOT* OR COMPUTER* OR DIGITAL* OR "E COUNSELING" OR ECOUNSELING OR E-COUNSELING OR "E COUNSELLING" OR ECOUNSELLING OR E-COUNSELLING OR "E HEALTH" OR EHEALTH OR E-HEALTH OR ELECTRONIC* OR EMAIL* OR E-MAIL* OR FACEBOOK OR ICT OR INTERNET OR MESSENGER OR MHEALTH OR M-HEALTH OR "MOBILE DEVICE" OR "MOBILE HEALTH" OR "MOBILE PHONE" OR "MOBILE-HEALTH" OR ONLINE* OR PHONE OR REMOTE OR SKYPE OR SMARTPHONE* OR SMS OR "SOCIAL MEDIA" OR TABLET OR TABLETS OR TECHNOLOG* OR "TELE HOME CARE" OR "TELE MEDICINE" OR TELECARE OR TELE-CARE OR TELECARING OR TELE-CARING OR TELECOMMUNICATION OR TELECONFERENCE* OR TELE-CONFERENCE* OR TELECONSULTATION* OR TELE-CONSULTATION* OR TELEHEALTH OR TELE-HEALTH OR TELEMEDICINE OR TELENURSE OR TELENURSING OR TELEPHONE OR TELEPHONES OR TELEPRESENCE OR TELEPRESENT OR TELEREHABILITATION OR TELE-REHABILITATION OR TELETHERAP* OR TELE-THERAP* OR TELETREATMENT OR TELE-TREATMENT OR VIDEOCONFERENCE OR VIDEOCONFERENCES OR VIRTUAL OR "WEB BASED" OR WEBBASED OR WEB-BASED OR "WORLD WIDE WEB").TI,AB,MP. OR EXP TELEMEDICINE/ OR EXP TECHNOLOGY/ OR EXP VIRTUAL REALITY/ OR EXP TELECONFERENCING/ OR EXP ONLINE THERAPY/ OR EXP TELEREHABILITATION/ OR EXP SOCIAL MEDIA/ OR EXP MOBILE DEVICES/	506844
3	(ADVICE* OR ADVISE* OR ADVISING OR COACH* OR CONSULT* OR COUNSELLING OR COUNSELING OR INSTRUCT* OR GUIDANCE* OR GUIDE* OR MENTORING OR "PASTORAL CARE" OR "SOCIAL SUPPORT*" OR SUPERVIS* OR "SUPPORT SERVICE*" OR "TALKING THERAP*").TI,AB,MP. OR EXP COUNSELING/ OR EXP SOCIAL SUPPORT/	700791
4	1 AND 2 AND 3	1151
5	(SURGICAL OR SURGERY OR GENETI* OR PROTEOMIC* OR GENOMIC* OR PROTEIN* OR NEUROIMAG*).TI,AB,MP.	285132
6	4 NOT 5	994

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	16
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Na
Support:			
Sources	5a	Indicate sources of financial or other support for the review	16
Sponsor	5b	Provide name for the review funder and/or sponsor	16
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	16
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4ff.
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7ff.
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	9f.
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	10f.
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	10f.
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	11
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	11
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	11
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	11
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	11ff.
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	11ff.
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	11ff.
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	11ff.
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	14
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	14

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-054157.R2
Article Type:	Protocol
Date Submitted by the Author:	13-Nov-2021
Complete List of Authors:	Hirt, Julian; Martin Luther University Halle Wittenberg, Institute for Health and Nursing Science; Eastern Switzerland University of Applied Sciences - Campus St Gallen, Center for Dementia Care, Institute of Applied Nursing Sciences, Department of Health Langer, Gero; Martin Luther University Halle Wittenberg, Institute for Health and Nursing Science Wilde, Fabian; Martin Luther University Halle Wittenberg, Institute for Health and Nursing Science Bauernschmidt, Dorothee; Martin Luther University Halle Wittenberg, Institute for Health and Nursing Science Meyer, Gabriele; Martin Luther University Halle Wittenberg, Institute for Health and Nursing Science Bieber, Anja; Martin Luther University Halle Wittenberg, Institute for Health and Nursing Science
Primary Subject Heading:	Nursing
Secondary Subject Heading:	Geriatric medicine
Keywords:	Dementia < NEUROLOGY, Information technology < BIOTECHNOLOGY & BIOINFORMATICS, GERIATRIC MEDICINE

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Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis

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ABSTRACT

Introduction

Information technology can enhance timely and individual support for people with dementia and informal carers. There is some evidence that people with dementia and informal carers would benefit from technology-based counselling. However, it remains unclear which features of those interventions relate to beneficial outcomes and which aspects are necessary for a successful implementation. Therefore, the primary objectives are: (i) to identify conditions of successful implementation of technology-based counselling interventions in dementia and (ii) to investigate the effectiveness of those interventions.

Methods and analysis

We will conduct a mixed-methods systematic review. The first primary objective requires evidence from various study designs addressing aspects on effective and non-effective implementation of technology-based counselling. This could be telephone-, web- or mobile-based interventions for people with dementia and informal carers. For the second primary objective, randomised controlled trials (RCTs) dealing with any outcomes will be included. Year of publication and language will not be restricted. We will search CINAHL, Cochrane Library, MEDLINE, PsycINFO, and Web of Science up to April 2021. Additionally, we will perform web searching and citation tracking. To achieve the first primary objective, a Qualitative Comparative Analysis (QCA) will be conducted. The QCA enables us to identify necessary or sufficient components for a successful implementation. To reach the second primary objective, a meta-analysis will be performed with respect to potential clinical and statistical heterogeneity of RCTs. The revised Risk of Bias tool RoB 2.0 will be used to check the risk of bias in RCTs. For all other study designs, the Mixed Method Appraisal Tool will be used.

Ethics and dissemination

Ethics approval is not required for this review. We will disseminate our findings through scientific and non-scientific journal articles and conference presentations as well as formats directed to the public and decision-makers in health care.

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Registration details

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Strengths and limitations of this study

- One methodological strength of our review is the combined determination of effectiveness of technology-based counselling interventions and of successful implementation features of such interventions.
- A comprehensive literature search combining electronic key database search and supplementary search methods with no restrictions on publication date and outcomes is another methodological strength of our review.
- Qualitative Comparative Analysis might uncover effective and non-effective combination(s) of intervention components that might be seen as a methodological strength of our review.
- Stakeholder participation and involvement in the Qualitative Comparative Analysis, the determination of the outcomes of interest for the GRADE approach as well as the interpretation of the review findings enables us to adopt the end-user perspectives and to strengthen the external validity of the review results.
- Since we expect only a few studies about technology-based counselling, this could result in limited data for the Qualitative Comparative Analysis.

Keywords

Dementia, Technology-based counselling, Counselling, Implementation, Qualitative Comparative Analysis

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INTRODUCTION

Description of the condition

Dementia is one of the major causes of disability and dependency among older people and is characterized by a progressive decline of cognitive functioning alongside other impairments, such as memory, executive functions, attention, emotional control, and language.[1] Nearly 50 million people have dementia worldwide, of whom approximately 60% live in low- and middle-income countries.[2] The number of new cases of dementia each year is estimated at over 9.9 million worldwide.[3] The average survival time after diagnosis seems to depend on the type of dementia; however, the length of survival is not related to age at diagnosis.[4] Results of a meta-analysis demonstrate an average survival time of 4.11 years after the diagnosis of Lewy body dementia (SD±4.10). People with a diagnosis of Alzheimer's disease have an average survival time of 5.66 years (SD±5.32).[5]

Common consequences of dementia are unmet needs, which can cause behavioural and psychological symptoms.[6] Hyperactivity and apathy show high persistence and incidence; depression and anxiety show low or moderate persistence and moderate incidence, and psychotic symptoms show low persistence and a moderate or low incidence. Around half of all the people with dementia area in need of assistance with personal care. According to an estimation by Prince et al., informal carers spent an average of 14 hours weekly supporting the person with dementia in activities of daily living (ADLs), such as dressing or bathing. Additionally, another 43 hours for instrumental activities of daily living (IADLs), for instance organisation of care or supervision, are needed. No significant differences between countries were identified.[7]

The majority of people with dementia live at home and receive care from a relative, mainly a spouse.[8] Informal carers of people with dementia seem to be less healthy and might experience more difficulties and stress than informal carers of people with other chronic conditions.[9 10] Three dimensions of burden for informal carers have been described: direct impact of caregiving upon informal carers' life (e.g., they lost control of their life or their health suffered), guilt (e.g., they should do more or could do a better job), and frustration or embarrassment (e.g., care recipient asks for too much help or carer is embarrassed or angered by the care recipient).[11] Nevertheless, providing care does not have a purely negative connotation, because informal carers feel vindicated in their role or have developed coping strategies.[12] Excessive informal carers' burden increases the risk of people with dementia having to move from their own home to a nursing

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home.[13] The worldwide COVID-19 (SARS-CoV-2) pandemic raised great concerns for people with dementia that were related to a lack of information, difficulties in safeguarding procedures or social isolation.[14] Supporting both people with dementia and their informal carers is essential in order to overcome anxiety, stress, agitation or withdrawal.[15]

Description of the intervention

Information technologies provide supportive options without the risk of infection. Beyond the current pandemic, the intensity of dementia care and the lack of available support services or existing barriers to get access – especially in rural areas – require additional and alternative support possibilities.[16] Information technology can provide timely and individual support for people with dementia and informal carers.[17-23] Various types of support are needed, such as information, education or counselling.[24] Counselling is defined as a conversation therapy that involves a trained therapist listening to individuals and helping them to find ways to cope with an emotional issue. It is differentiated from information, education, instruction and case management.[25] Few studies have investigated the effects of counselling for people with dementia and their informal carers.[26] Potential effects include changes in the attitudes of couples confronted with Alzheimer's disease towards becoming more acceptant of the diagnosis along with a more optimistic view for the future.[27]

How the intervention might work

Counselling can be provided through a face-to-face interaction, in a group, or through the use of technological channels, namely: the phone, e-mail, or live chat services. People who received telephone counselling believed that the intervention was attractive with regard to convenience, accessibility, control or inhibition.[28] A recent mixed-methods study uncovered the needs and preferences of informal carers of people with dementia and shows that an online intervention might be an adequate format to provide individual support. From their perspective, tailored programs that are simple to use and interaction with a professional within the online intervention are highly needed.[29] A recent scoping review on virtual support groups for informal carers of people with dementia also highlights the need for tailored, easy-accessible, and moderated online support.[30] Therefore, online counselling might meet the needs of many clients and

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result in positive outcomes.[31] For example, virtual appointments could reduce stress among carers because counselling sessions can be carried out from home. This is in particular of interest, if carers have a problem leaving their home where they care for a person with dementia. Online counselling can be differentiated from other online services (e.g., web-based education or therapeutic interventions, therapeutic software or online activities, which are not stand-alone services).[32]

Alternatives to face-to-face interventions are crucial in the current COVID-19 pandemic and furthermore have the potential for a more timely and individualised support in dementia. Few systematic reviews investigated technology-based interventions for people with dementia and informal carers (e.g., telephone counselling,[33] or support interventions via computer and telephone [34]). The review by Lins et al. included nine randomised controlled trials (RCTs) and two qualitative studies.[33] The mixed-method review aimed to identify the efficacy of telephone counselling for informal carers of people with dementia and asked for the experiences of informal carers with the intervention. The results demonstrate moderate evidence that telephone counselling without other intervention components can reduce depressive symptoms in informal carers. The experiences of informal carers included barriers and facilitators for successful implementation, counsellor's emotional attitude, and content of the telephone counselling. The authors concluded that the efficacy of the intervention was influenced by a lack of knowledge related to the aspects of telephone counselling . The review by Waller et al. included studies related to interventions delivered by a computer (n=10), multiple modalities (n=9), or telephone (n=15).[34] Due to the expanding number of related publications, one could assume an increasing demand for better support for informal carers of people with dementia. In conclusion, a need for support and advice for the person with dementia and the informal carer as a dyad was identified.

An overview of systematic reviews investigated psychosocial interventions in dementia and identified counselling and psychotherapy as the second largest category of interventions for informal carers, showing some evidence confirming the effectiveness of these interventions.[35] Technology-based interventions were also effective, but components or aims of these interventions were not described. As in the case of the reviews by Lins et al. and Waller et al., conditions of telephone-based counselling or technology-based interventions were not described in detail.[33 34]

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An update and extension of the review by Lins et al. is timely and warranted.[33] In addition to telephone-based counselling, we will cover other technologies such as computer or web-based interventions. Our review itself will address both the target group of informal carers of people with dementia and studies that report on interventions for people with dementia themselves. That step would fulfil the global vision of evidence-based care in dementia, providing care and support for the needs of people with dementia and their informal carers to enhance their dignity, respect, autonomy and equality.[36]

Objectives

Although there is some evidence that people with dementia and informal carers might benefit from technology-based counselling [33 34], it remains unclear what features of those interventions relate to successful outcomes and what aspects of implementation are necessary and successful. Our review will identify conditions of a successful implementation of technology-based counselling interventions in dementia and investigate the effectiveness of those interventions. The results will present the effectiveness, i.e., beneficial impact and potential adverse events of technology-based counselling for people with dementia and their carers. This is necessary to guide clinical practice concerning the use of such interventions. A comprehensive overview of successful and non-successful intervention components will facilitate a structured guidance for clinical implementation. The objectives of our review are therefore (i) to identify conditions that are aligned with successful implementation of technology-based counselling interventions in dementia (i.e., implementation facilitators and barriers) and (ii) to investigate the effectiveness of these interventions. The following two review questions are raised:

- What are necessary or sufficient conditions that are aligned with successful implementation and effective outcomes of technology-based counselling interventions for people with dementia and their carers?
- How effective are technology-based counselling interventions for people with dementia and their carers?

METHODS AND ANALYSIS

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We will conduct a mixed-methods systematic review using (i) the methodological approach of Qualitative Comparative Analysis (QCA), aimed at identifying the features of or conditions for successful interventions,[37] and (ii) a meta-analysis, aimed at determining the overall effectiveness of technology-based counselling in dementia. PRISMA-P was used to structure this systematic review protocol.[38]

Eligibility criteria

To reach our review objectives, we are interested in all information concerning the development, evaluation, and implementation of technology-based counselling interventions in dementia. Therefore, we defined the following eligibility criteria.

Populations

We will include studies on interventions for people with dementia and/or their informal carers. There will be no restrictions concerning setting, gender, age, ethnic background, type and severity of dementia, or relationship between people with dementia and their informal carers. Studies on people with mild cognitive impairment or their informal carers will be excluded. Studies with a mixed sample of informal carers of people with dementia and other conditions will be included if the results for the dementia subgroup have been reported separately. We will also exclude studies on interventions for health and social care professionals.

Interventions

We will include tailored technology-based counselling interventions for people with dementia and/or their informal carers. Technology-based interventions use technologies such as telephone, computer, and web- or mobile-based applications to enable the delivery of the intervention instantly and/or remotely. Counselling is a complex intervention that might consist of different intervention components or be part of multi-component interventions. To assess the impact of counselling, we will focus on interventions with a core component of counselling. Thus, we will include tailored counselling interventions provided by health and social care professionals such as nurses, physicians or psychologists aiming to support people with dementia and/or their informal carers (e.g., focusing management of daily living, disease and care

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management as well as functioning and participation). Therefore, the interventions of interest will be technology-based, provided human-to-human, but may be asynchronous. Studies with interventions that might use counselling as a component of complex care programs and mainly focus on coordination of care such as case management will be excluded. We will also exclude specific and standardised interventions such as cognitive behaviour therapy, genetic counselling as well as counselling with regard to predictive diagnostics, screening for dementia or diagnostic procedures. Counselling interventions provided by non-professionals (i.e. non-paid) will be excluded.

Comparators

Technology-based counselling will be compared to no treatment, to standard or usual care, and to in presence or face-to-face counselling. Comparisons between various technology-based interventions will also be considered for inclusion.

Outcomes

We will include all features of effective and non-effective intervention implementation and there will be no restrictions regarding outcomes on the effectiveness of interventions. Our primary outcome is defined as the impact of technology-based counselling on depressive symptoms experienced by people with dementia and their carers. As secondary outcomes, we will consider outcomes on the (i) patient and carer levels (such as burden, well-being, delay or avoidance of transition to a nursing home, decision-making, and acceptance of as well as satisfaction with technology-based counselling) and on the (ii) healthcare system level (such as usage and costs of technology-based counselling).[34 35] We do not expect a negative impact from counselling interventions for people with dementia and carers. However, adverse events will be included.

Design of primary studies

The first primary objective requires evidence from various study designs addressing aspects of implementation success such as qualitative studies, descriptive quantitative studies (e.g., before-after studies, clinical controlled studies, cross-sectional studies, surveys), and mixed-methods research.

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To meet the second primary objective, we will include individual and cluster parallel-group RCTs, also RCTs using cross-over and stepped-wedge designs.

Information sources

We will perform a comprehensive literature search combining an electronic database search with supplementary search methods and with no limitations regarding language and year of publication.[39 40]

Electronic database search

We will search CINAHL, Cochrane Library including CENTRAL (Cochrane Central Register of Controlled Trials), MEDLINE via PubMed, PsycINFO via Ovid, and the Web of Science Core Collection up to April 2021 with no use of filters and limiters.

Supplementary search methods

Google Scholar will be used for free web searching with highly relevant free text terms in order to identify further studies and conference abstracts not yet identified as well as grey literature, thus minimizing publication bias. We will search further primary studies using (i) hand searching of revealed reviews and (ii) forward and backward citation tracking of included studies using Scopus. To identify ongoing studies, we will search the International Clinical Trials Registry Platform (ICTRP) via CENTRAL, a meta study registry provided by the World Health Organization covering national registries from 17 countries worldwide.

Search update

An update for potentially relevant studies will be conducted in electronic databases one year after the initial search. For the search update, we will use the same search methods and strategies.

Search strategy

We will use database-specific search strategies based on database-specific controlled vocabulary, for instance MeSH (MEDLINE), and unspecified free text terms. The search strategies will be developed and

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peer-reviewed within the review group using PRESS (Peer Review of Electronic Search Strategies). A researcher experienced in designing search strategies will support us during that process.[41] Our search strategy will contain three components (Dementia, Technology, and Counselling). Search terms and their synonyms will be identified using an earlier study by our review group,[33] an orienting search using MEDLINE via PubMed, and through the experience of the review group members. Additionally, text mining of titles and abstracts of previously known studies will be applied to identify relevant text terms used in the core literature. Based on the identified search terms, we will look for further search terms with the help of a thesaurus [42] and by checking entry terms given in the MeSH browser. For unspecified free text terms, we will search title, abstract, and keyword fields, if available. Controlled vocabulary will be used in the corresponding fields and additionally in the title, abstract, and keyword search fields in order to cover publications not yet tagged with controlled vocabulary.[43] The final search strategy per database is provided in the supplementary file.

Study records

Two reviewers will independently screen the identified titles, abstracts, and, if eligible, full texts for inclusion and exclusion using the Rayyan web app.[44] We will discuss conflicting results within the screening team and with a third member of the review group - if necessary. Two reviewers will develop and pilot a data extraction sheet using ten percent of the included studies. Subsequently, one reviewer will extract data. A second reviewer will check the data extraction for its accuracy. We will discuss conflicting results within the data extracting team and, if necessary, also with a third member of the review group.

Data items

We will extract data on study characteristics (e.g., study design, country), participants' characteristics (e.g., age, gender, type and severity of dementia), intervention and control group characteristics (e.g., type, format, length, frequency), outcome(s) or phenomena of interest (e.g., implementation facilitators and barriers), measurement(s), and results. In addition, we will extract data according to the criteria of TIDieR and CReDECI-2 to provide a description of the reported intervention characteristics and their theoretical background.[45 46]

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Outcomes and prioritization

We will involve our advisory board in determining the outcomes of interest following the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach (for details see below).[47]

Risk of bias and critical appraisal in individual studies

We will assess the risk of bias in RCTs using the revised risk of bias tool (RoB 2.0).[48] For all other study designs, we will use the Mixed Method Appraisal Tool, a comprehensive critical appraisal tool provided by Hong et al.[49] All appraisals will be conducted by two independent reviewers. We will discuss conflicting results within the appraisal team and, if necessary, with a third member of the review group.

Data synthesis

Qualitative Comparative Analysis (QCA)

To meet our first primary objective, we will conduct a QCA to identify conditions of interventions that are related to successful implementation and effective outcomes. Key elements of the QCA are the configurations of the conditions. These are different scenarios of intervention components and potential facilitators as well as barriers of their implementation. The QCA enables us to identify necessary or sufficient components for implementation aspects and successful interventions. QCA contains the steps which are displayed in the following sub-chapters.[37]

Building the data table using selected conditions and outcomes

Unlike systematic reviews on effectiveness interventions, the QCA approach considers various study designs. Studies ("cases" in the QCA terminology) will be added or removed in an iterative process until a later stage of the research.

The selection and definition of the conditions and outcomes is based on the preliminary findings from the literature.[50] The conditions will be grouped into different domains such as setting and participants' characteristics, intervention characteristics, or planned and unplanned measures to facilitate

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implementation. To structure our conditions and domains, we will develop a logic model. The number of conditions is expected to be relatively low, and for each condition a clear hypothesis will be formulated according to its expected relation to the outcome.[37] Identified conditions and outcomes will be mapped in a raw data table and calibrated towards their so-called set membership. To prepare the calibration process, a coding framework is needed to provide instructions and values for the data extraction. For each case (= included study), the conditions and outcomes will be determined in crisp-sets with dichotomous values of 0 or 1, or with fuzzy-sets for degrees of membership, transformed in values between 0 and 1.

Constructing a truth table and resolving contradictory configurations

In a next step, the possible combinations of conditions will be tabulated. Each line in the table represents a configuration of possible conditions using logical operators. The number of configurations will be 2*k, where "k" represents the number of conditions. The configuration represents a potential set where studies can be members or non-members. Studies showing the same configuration will be included in one set and studies showing different configurations will be included in other sets. Therefore, so-called truth tables using the dichotomous or transformed values of crisp- and fuzzy-sets will be developed when configurations (i) are supported by (one or more) cases, (ii) are counter-intuitive or showed an identical combination of conditions, (iii) occurred infrequently, and (iv) triggered positive and negative outcomes at the same time.

Boolean minimisation and consideration of the 'logical remainders' cases

We will subsequently use Boolean minimisation to identify configurations that are observed to adequately lead to a sufficient and specific outcome. We will then consider the plausibility of the configurations and search for consistent conclusions in individual studies (consideration of logical remainders cases). Identified configurations that are sufficiently specific can form different paths towards an outcome.[50] Therefore, coefficients of coverage will be expressed, and the theoretical relevance will be considered. Furthermore, a "return to the cases" step will help us to evaluate how these results are represented in a study and across studies.[47 51]

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Interpretation of the results

The interpreted QCA results will be used for the development of a logical model that can support the practice of technology-based counselling. Furthermore, the newly introduced logical model has the potential to sufficiently inform further research activities in order to compromise inadequate knowledge about the effective steps of implementation. We will use an appropriate software for that analytic step.

Meta-Analysis

To meet our second primary objective, we will perform a meta-analysis with respect to the potential clinical and statistical heterogeneity of RCTs. Clinical heterogeneity will be indicated by the comparability of interventions based on intervention characteristics and components.[52] Groupings consisting of studies with similar characteristics and interventions might reach comparability. Therefore, it will be possible to conduct stratified meta-analyses given a sufficient number of studies across study groups. We will evaluate the statistical heterogeneity using the I^2 method.[53] A value higher than 50% will be regarded as significantly heterogeneous. Here, we will use the random-effects model and in the case of non-significant heterogeneity, we will use the fixed-effects model to calculate the pooled effect sizes.[54] A p value less than 0.05 will be considered as statistically significant. For meta-analysis of dichotomous results, we will calculate the relative risks (RR) with CI 95%. For the meta-analysis of continuous results, we will use the mean difference (MD) and CI 95%. If different outcome measurements have been used, we will use the standardized mean difference (SMD) and CI 95%. To assess the influence of different conditions on the effects of technology-based counselling (e.g., setting, recipient, facilitator, and type of the intervention), we will conduct sensitivity analyses based on subgroup meta-analyses.[55] We will use the software "R" in its current version for our calculations. The results of the meta-analysis will be presented in forest plots.[56]

If the previously described assessments indicate heterogeneous interventions and study results, we will provide a narrative synthesis based on tabular data extraction. Additionally, we will summarise effects, risk of bias, intervention and study characteristics graphically by means of a Harvest plot to report appropriately on the complexity and heterogeneity of the included studies.[57 58]

Meta-bias

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Given a sufficient number of included studies to meet our second primary objective, risk for publication bias will be assessed graphically using a funnel plot complemented by a statistical test.[59]

Confidence in cumulative evidence

The quality of evidence will be assessed using the GRADE approach by rating the quality of evidence as high, moderate, low or very low.[60] The combined results of meta-analyses and the GRADE approach will be presented in a summary of findings table.[61]

Patient and Public Involvement

We will include representatives of people with dementia and informal carers, i.e. people engaged in a support organisation that supports dementia care, and an informal carer of a person with dementia. The advisory board will consist of two or three people. Recommendations by the advisory board will be used for the qualitative steps of the QCA, where experience and knowledge of the subject are needed for the identification and the assessment of conditions that are important for the intervention. Furthermore, the advisory board will be involved in determining the outcomes of interest for the GRADE approach as well as the interpretation of the review findings.[47]

ETHICS AND DISSEMINATION

Ethics approval is not required for this review protocol. The members of the advisory board participate voluntarily in the review process. Ongoing informed consent will be obtained.

Our dissemination strategy uses multiple channels of distribution to share our review work and results with academic and non-academic stakeholders, the society, and interested individuals.[62]

The final review will be published in an international open access journal relevant in the field of dementia care. Additionally, we will discuss our review results with experts at national and international conferences. Plain language summaries of the review will be placed on our institutional websites and social media profiles (e.g., Twitter, ResearchGate). A second review report will be published in a non-academic journal to share our review results with other interested groups such as practitioners and carers of people with dementia. Furthermore, we will participate in interactive mass media platforms to enable interested groups

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to ask questions. A cartoon video will be accessible and disseminated online. Additionally, the results will be printed in newsletters of organisations representing people with dementia as well as patient advocacy groups such as Deutsche Alzheimer Gesellschaft, Alzheimer Schweiz, Österreichische Alzheimer Gesellschaft, and Alzheimer Europe. Furthermore, we will contact insurance companies and other stakeholders to disseminate our research results.

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Authors' contributions

Substantial contributions to the conception or design of the work: JH, GL, GM, AB

Drafting the work: JH, AB

Revising it critically for important intellectual content: JH, GL, FW, DB, GM, AB

Final approval of the version to be published: JH, GL, FW, DB, GM, AB

Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy and integrity of any part of the work are appropriately investigated and resolved: JH, GL, FW, DB, GM, AB

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Competing interests' statement

All the authors declare that there is no competing interest.

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REFERENCES

1. WHO. ICD-11 for Mortality and Morbidity Statistics. Neurocognitive disorders 2019 [Available from: https://www.who.int/classifications/icd/en/.
2. WHO. Dementia, 2019.
3. Prince M, Wimo A, Guerchet M, et al. World Alzheimer Report 2015: The Global Impact of Dementia: An Analysis of Prevalence, Incidence, Cost and Trend London2015 [Available from: https://www.alz.co.uk/research/WorldAlzheimerReport2015.pdf accessed 19.03.2020.
4. Brodaty H, Seeher K, Gibson L. Dementia time to death: a systematic literature review on survival time and years of life lost in people with dementia. *Int Psychogeriatr* 2012;24(7):1034-45. doi: 10.1017/S1041610211002924 [published Online First: 2012/02/14]
5. Mueller C, Soysal P, Rongve A, et al. Survival time and differences between dementia with Lewy bodies and Alzheimer's disease following diagnosis: A meta-analysis of longitudinal studies. *Ageing Res Rev* 2019;50:72-80. doi: 10.1016/j.arr.2019.01.005 [published Online First: 2019/01/10]
6. Hung L, Lee PA, Au-Yeung AT, et al. Adopting a Clinical Assessment Framework in Older Adult Mental Health. *J Psychosoc Nurs Ment Health Serv* 2016;54(7):26–31. doi: 10.3928/02793695-20160616-05
7. Prince M, Prina M, Guerchet M. World Alzheimer Report 2013: Journey of Caring: An analysis of long-term care for dementia London2013 [Available from: https://www.alz.co.uk/research/WorldAlzheimerReport2013.pdf accessed 19.03.2020.
8. Wimo A, Gauthier S, Prince M. Global estimates of informal care 2018 [Available from: https://www.alz.co.uk/adi/pdf/global-estimates-of-informal-care.pdf accessed 15.05.2020.
9. Schoenmakers B, Buntinx F, Delepeleire J. Factors determining the impact of care-giving on caregivers of elderly patients with dementia. A systematic literature review. *Maturitas* 2010;66(2):191-200. doi: 10.1016/j.maturitas.2010.02.009 [published Online First: 2010/03/24]
10. Isik AT, Soysal P, Solmi M, et al. Bidirectional relationship between caregiver burden and neuropsychiatric symptoms in patients with Alzheimer's disease: A narrative review. *Int J Geriatr Psychiatry* 2019;34(9):1326-34. doi: 10.1002/gps.4965 [published Online First: 2018/09/11]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

11. Springate BA, Tremont G. Dimensions of caregiver burden in dementia: impact of demographic, mood, and care recipient variables. *Am J Geriatr Psychiatry* 2014;22(3):294-300. doi: 10.1016/j.jagp.2012.09.006 [published Online First: 2013/04/10]

12. Lloyd J, Patterson T, Muers J. The positive aspects of caregiving in dementia: A critical review of the qualitative literature. *Dementia* 2016;15(6):1534-61. doi: 10.1177/1471301214564792 [published Online First: 2016/10/30]

13. Cepoiu-Martin M, Tam-Tham H, Patten S, et al. Predictors of long-term care placement in persons with dementia: a systematic review and meta-analysis. *Int J Geriatr Psychiatry* 2016;31(11):1151-71. doi: 10.1002/gps.4449 [published Online First: 2016/10/19]

14. Wang H, Li T, Barbarino P, et al. Dementia care during COVID-19. *The Lancet* 2020;395(10231):1190-91. doi: 10.1016/s0140-6736(20)30755-8

15. WHO. ADI offers advice and support during COVID-19 https://www.alz.co.uk/news/adi-offers-advice-and-support-during-covid-192020 [Available from: https://www.alz.co.uk/news/adi-offers-advice-and-support-during-covid-19 accessed 05.05.2020.

16. Innes A, Morgan D, Kosteniuk J. Dementia care in rural and remote settings: a systematic review of informal/family caregiving. *Maturitas* 2011;68(1):34-46. doi: 10.1016/j.maturitas.2010.10.002 [published Online First: 2010/11/26]

17. Bejan A, Gündogdu R, Butz K, et al. Using multimedia information and communication technology (ICT) to provide added value to reminiscence therapy for people with dementia: Lessons learned from three field studies. *Z Gerontol Geriatr* 2017;51(1):9–15. doi: 10.1007/s00391-017-1347-7

18. D'Onofrio G, Sancarlo D, Ricciardi F, et al. Information and Communication Technologies for the Activities of Daily Living in Older Patients with Dementia: A Systematic Review. *J Alzheimer's Dis* 2017;Article in Press doi: 10.3233/JAD-161145

19. Lauriks S, Reinersmann A, van der Roest HG, et al. Review of ICT-based services for identified unmet needs in people with dementia. *Ageing Res Rev* 2007;6(3):223–46. doi: 10.1016/j.arr.2007.07.002

20. Lucero RJ, Fehlberg EA, Patel AGM, et al. The effects of information and communication technologies on informal caregivers of persons living with dementia: A systematic review. *Alzheimers Dement* 2019;5:1–12. doi: 10.1016/j.trci.2018.11.003

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

21. McHugh JE, Wherton JP, Prendergast DK, et al. Identifying opportunities for supporting caregivers of persons with dementia through information and communication technology. *Gerontechnology* 2013;10(4) doi: 10.4017/gt.2012.10.4.003.00

22. Noone C, McSharry J, Smalle M, et al. Video calls for reducing social isolation and loneliness in older people: a rapid review. *Cochrane Database Syst Rev* 2020;5:CD013632. doi: 10.1002/14651858.CD013632

23. Pleasant M, Molinari V, Dobbs D, et al. Effectiveness of online dementia caregivers training programs: A systematic review. *Geriatr Nurs* 2020;Article in Press doi: 10.1016/j.gerinurse.2020.07.004

24. Whitlatch CJ, Orsulic-Jeras S. Meeting the Informational, Educational, and Psychosocial Support Needs of Persons Living With Dementia and Their Family Caregivers. *Gerontologist* 2018;58(suppl_1):S58-S73. doi: 10.1093/geront/gnx162 [published Online First: 2018/01/24]

25. NHS. Counselling https://www.nhs.uk/conditions/counselling/2017 [accessed 17.1.2020.

26. Dickinson C, Dow J, Gibson G, et al. Psychosocial intervention for carers of people with dementia: What components are most effective and when? A systematic review of systematic reviews. *Int Psychogeriatr* 2017;29(1):31-43. doi: 10.1017/S1041610216001447 [published Online First: 2016/09/27]

27. Elvish R, Lever S-J, Johnstone J, et al. Psychological interventions for carers of people with dementia: A systematic review of quantitative and qualitative evidence. *Counsell Psychother Res J* 2013;13(2):106-25. doi: 10.1080/14733145.2012.739632

28. Reese RJ, Conoley C, W., Brossart DF. The Attractiveness of Telephone Counseling: An Empirical Investigation of Client Perceptions. *J Couns Dev* 2006;84:54-60.

29. Teles S, Paúl C, Sosa Napolskij M, et al. Dementia caregivers training needs and preferences for online interventions: a mixed-methods study. *J Clin Nurs* 2020;Article in Press doi: 10.1111/jocn.15445

30. Armstrong MJ, Alliance S. Virtual Support Groups for Informal Caregivers of Individuals With Dementia: A Scoping Review. *Alzheimer Dis Assoc Disord* 2019;33(4):362–69. doi: 10.1097/WAD.0000000000000349

31. Harris B, Birnbaum R. Ethical and Legal Implications on the Use of Technology in Counselling. *Clin Soc Work J* 2015;43:133–41.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

32. Dowling M, Rickwood D. Online Counseling and Therapy for Mental Health Problems: A Systematic Review of Individual Synchronous Interventions Using Chat. *J Tech Hum Serv* 2013;31(1):1-21.

33. Lins S, Hayder-Beichel D, Ruecker G, et al. Efficacy and experiences of telephone counselling for informal carers of people with dementia. *Cochrane Database Syst Rev* 2014(9):CD009126. doi: 10.1002/14651858.CD009126.pub2

34. Waller A, Dilworth S, Mansfield E, et al. Computer and telephone delivered interventions to support caregivers of people with dementia: a systematic review of research output and quality. *BMC Geriatr* 2017;17(1):265. doi: 10.1186/s12877-017-0654-6 [published Online First: 2017/11/18]

35. Clarkson P, Hughes J, Xie C, et al. Overview of systematic reviews: Effective home support in dementia care, components and impacts-Stage 1, psychosocial interventions for dementia. *J Adv Nurs* 2017;73(12):2845-63. doi: 10.1111/jan.13362

36. WHO. Global action plan on the public health response to dementia 2017 - 2025. 2017

37. Rihoux B, Ragin CC, editors. *Configurational comparative methods: Qualitative comparative analysis (QCA) and related techniques*. Los Angeles: SAGE, 2009.

38. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1. doi: 10.1186/2046-4053-4-1

39. Cooper C, Booth A, Varley-Campbell J, et al. Defining the process to literature searching in systematic reviews: a literature review of guidance and supporting studies. *BMC Med Res Methodol* 2018;18:85. doi: 10.1186/s12874-018-0545-3

40. Lefebvre C, Glanville J, Briscoe S, et al. Searching for and selecting studies. In: Higgins JPT, Thomas J, eds. Cochrane Handbook for Systematic Reviews of Interventions. 2 ed. Hoboken: Wiley Online Library 2019:67–108.

41. McGowan J, Sampson M, Salzwedel DM, et al. PRESS Peer Review of Electronic Search Strategies: 2015 Guideline Statement. *J Clin Epidemiol* 2016;75:40–46.

42. Dictionary.com. Thesaurus.com 2020 [Available from: http://www.thesaurus.com/ accessed 28.04.2020.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

43. Rodriguez RW. Comparison of indexing times among articles from medical, nursing, and pharmacy journals. *Am J Health Syst Pharm* 2016;73(8):569–75. doi: 10.2146/ajhp150319

44. Ouzzani M, Hammady H, Fedorowicz Z, et al. Rayyan-a web and mobile app for systematic reviews. *Syst Rev* 2016;5:210. doi: 10.1186/s13643-016-0384-4

45. Hoffmann TC, Glasziou P, Boutron I, et al. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *BMJ* 2014;348:g1687.

46. Möhler R, Köpke S, Meyer G. Criteria for Reporting the Development and Evaluation of Complex Interventions in healthcare: revised guideline (CReDECI 2). *Trials* 2015;16(1):204.

47. Langer G, Meerpohl JJ, Perleth M, et al. GRADE-Leitlinien: 2. Formulierung der Fragestellung und Entscheidung über wichtige Endpunkte. *Z Evid Fortbild Qual Gesundhwes* 2012;106(5):369–76.

48. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;2:l4898. doi: 10.1136/bmj.l4898

49. Hong QN, Fàbregues S, Bartlett G, et al. The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. *Education for Information* 2018;34(4):285–91. doi: 10.3233/EFI-180221

50. Schneider CQ, Wagemann C. Standards of Good Practice in Qualitative Comparative Analysis (QCA) and Fuzzy-Sets. *Comp Sociol* 2010;9(3):397–418. doi: 10.1163/156913210X12493538729793

51. Kahwati L, Jacobs S, Kane H, et al. Using qualitative comparative analysis in a systematic review of a complex intervention. *Syst Rev* 2016;5:82. doi: 10.1186/s13643-016-0256-y

52. West SL, Gartlehner G, Mansfield AJ, et al. Comparative Effectiveness Review Methods: Clinical Heterogeneity. Agency for Healthcare Research and Quality. Rockville, 2010.

53. Higgins JPT, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557–60. doi: 10.1136/bmj.327.7414.557

54. Hedges LV, Vevea J. Fixed-and random-effects models in meta-analysis. *Psychol Methods* 1998;3(4):486–504.

55. Khan KS, Kunz R, Kleijnen J, et al. Systematic Reviews to Support Evidence Bsaed Medicine. How to Review and Apply Findings of Healthcare Research London: Hodder & Stoughton.

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56. Lewis S, Clarke M. Forest plots: trying to see the wood and the trees. *BMJ* 2001;322(7300):1479–80. doi: 10.1136/bmj.322.7300.1479

57. Higgins JPT, López-López JA, Becker BJ, et al. Synthesising quantitative evidence in systematic reviews of complex health interventions. *BMJ Glob Health* 2019;4(Suppl 1):e000858. doi: 10.1136/bmjgh-2018-000858

58. Ogilvie D, Fayter D, Petticrew M, et al. The harvest plot: a method for synthesising evidence about the differential effects of interventions. *BMC Med Res Methodol* 2008;8:1–8.

59. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;315(7109):629–34.

60. Schünemann HB, Jan; Guyatt, Gordon; Oxman, Andrew. GRADE Handbook: Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach2013.

61. Guyatt GH, Thorlund K, Oxman AD, et al. GRADE guidelines: 13. Preparing summary of findings tables and evidence profiles-continuous outcomes. *J Clin Epidemiol* 2013;66(2):173–83. doi: 10.1016/j.jclinepi.2012.08.001

62. Cochrane. Checklist and Guidance: For disseminating findings from Cochrane intervention reviews: Version 1.0. Project in collaboration with Cochrane Knowledge Translation and Cochrane Norway, 2019.

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Technology-based counselling in dementia (TeCoDem): study protocol of a mixed-methods systematic review with qualitative comparative analysis and meta-analysis

Supplement: search strategies

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2	"APP"[TITLE/ABSTRACT] OR "APPS"[TITLE/ABSTRACT] OR "AUGMENTED REALIT*"[TITLE/ABSTRACT] OR "BLENDED TREATMENT"[TITLE/ABSTRACT] OR "BLENDED THERAPY"[TITLE/ABSTRACT] OR "BLOG"[TITLE/ABSTRACT] OR "CELL PHONE*"[TITLE/ABSTRACT] OR "CELLPHONE*"[TITLE/ABSTRACT] OR "CHAT"[TITLE/ABSTRACT] OR "CHATS"[TITLE/ABSTRACT] OR "CHATBOT*"[TITLE/ABSTRACT] OR "COMPUTER*"[TITLE/ABSTRACT] OR "DIGITAL*"[TITLE/ABSTRACT] OR "E COUNSELING"[TITLE/ABSTRACT] OR "ECOUNSELING"[TITLE/ABSTRACT] OR "E COUNSELING"[TITLE/ABSTRACT] OR "E COUNSELLING"[TITLE/ABSTRACT] OR "ECOUNSELLING"[TITLE/ABSTRACT] OR "E COUNSELLING"[TITLE/ABSTRACT] OR "E HEALTH"[TITLE/ABSTRACT] OR "EHEALTH"[TITLE/ABSTRACT] OR "E HEALTH"[TITLE/ABSTRACT] OR "ELECTRONIC*"[TITLE/ABSTRACT] OR "EMAIL*"[TITLE/ABSTRACT] OR "E MAIL*"[TITLE/ABSTRACT] OR "FACEBOOK"[TITLE/ABSTRACT] OR "ICT"[TITLE/ABSTRACT] OR "INTERNET"[TITLE/ABSTRACT] OR "MESSENGER"[TITLE/ABSTRACT] OR "MHEALTH"[TITLE/ABSTRACT] OR "M-HEALTH"[TITLE/ABSTRACT] OR "MOBILE DEVICE"[TITLE/ABSTRACT] OR "MOBILE HEALTH"[TITLE/ABSTRACT] OR "MOBILE PHONE"[TITLE/ABSTRACT] OR "MOBILE HEALTH"[TITLE/ABSTRACT] OR "ONLINE*"[TITLE/ABSTRACT] OR "PHONE"[TITLE/ABSTRACT] OR "REMOTE"[TITLE/ABSTRACT] OR "SKYPE"[TITLE/ABSTRACT] OR "SMARTPHONE*"[TITLE/ABSTRACT] OR "SMS"[TITLE/ABSTRACT] OR "SOCIAL MEDIA"[TITLE/ABSTRACT] OR "TABLET"[TITLE/ABSTRACT] OR "TABLETS"[TITLE/ABSTRACT] OR "TECHNOLOG*"[TITLE/ABSTRACT] OR "TELE HOME CARE"[TITLE/ABSTRACT] OR "TELE MEDICINE"[TITLE/ABSTRACT] OR "TELECARE"[TITLE/ABSTRACT] OR "TELE-CARE"[TITLE/ABSTRACT] OR "TELECARING"[TITLE/ABSTRACT] OR "TELE-CARING"[TITLE/ABSTRACT] OR "TELECOMMUNICATION"[TITLE/ABSTRACT] OR "TELECONFERENCE*"[TITLE/ABSTRACT] OR "TELE CONFERENCE*"[TITLE/ABSTRACT] OR "TELECONSULTATION*"[TITLE/ABSTRACT] OR "TELE CONSULTATION*"[TITLE/ABSTRACT] OR "TELEHEALTH"[TITLE/ABSTRACT] OR "TELE-HEALTH"[TITLE/ABSTRACT] OR "TELEMEDICINE"[TITLE/ABSTRACT] OR "TELENURSE"[TITLE/ABSTRACT] OR "TELENURSING"[TITLE/ABSTRACT] OR "TELEPHONE"[TITLE/ABSTRACT] OR "TELEPHONES"[TITLE/ABSTRACT] OR "TELEPRESENCE"[TITLE/ABSTRACT] OR "TELEPRESENT"[TITLE/ABSTRACT] OR "TELEREHABILITATION"[TITLE/ABSTRACT] OR "TELE-REHABILITATION"[TITLE/ABSTRACT] OR "TELETHERAP*"[TITLE/ABSTRACT] OR "TELE THERAP*"[TITLE/ABSTRACT] OR "TELETREATMENT"[TITLE/ABSTRACT] OR "TELE-TREATMENT"[TITLE/ABSTRACT] OR "VIDEOCONFERENCE"[TITLE/ABSTRACT] OR "VIDEOCONFERENCES"[TITLE/ABSTRACT] OR "VIRTUAL"[TITLE/ABSTRACT] OR "WEB BASED"[TITLE/ABSTRACT] OR "WEBBASED"[TITLE/ABSTRACT] OR "WEB BASED"[TITLE/ABSTRACT] OR "WORLD WIDE WEB"[TITLE/ABSTRACT] OR "TELEMEDICINE"[MESH TERMS] OR "TELENURSING"[MESH TERMS] OR "TECHNOLOGY"[MESH TERMS] OR "VIRTUAL REALITY"[MESH TERMS] OR "TELECOMMUNICATIONS"[MESH TERMS] OR "VIDEOCONFERENCING"[MESH TERMS] OR "TELEREHABILITATION"[MESH TERMS] OR "INTERNET-BASED INTERVENTION"[MESH TERMS] OR "SOCIAL MEDIA"[MESH TERMS] OR "INFORMATION TECHNOLOGY"[MESH TERMS] OR "REMOTE CONSULTATION"[MESH TERMS] OR "DISTANCE COUNSELING"[MESH TERMS] OR "VIRTUAL REALITY"[MESH TERMS] OR "TELEPHONE"[MESH TERMS] OR "CELL PHONE"[MESH TERMS]	2123169
3	"ADVICE*"[TITLE/ABSTRACT] OR "ADVISE*"[TITLE/ABSTRACT] OR "ADVISING"[TITLE/ABSTRACT] OR "COACH*"[TITLE/ABSTRACT] OR "CONSULT*"[TITLE/ABSTRACT] OR "COUNSELLING"[TITLE/ABSTRACT] OR "COUNSELING"[TITLE/ABSTRACT] OR "INSTRUCT*"[TITLE/ABSTRACT] OR "GUIDANCE*"[TITLE/ABSTRACT] OR "GUIDE*"[TITLE/ABSTRACT] OR "MENTORING"[TITLE/ABSTRACT] OR "PASTORAL CARE"[TITLE/ABSTRACT] OR "SOCIAL SUPPORT*"[TITLE/ABSTRACT] OR "SUPERVIS*"[TITLE/ABSTRACT] OR "SUPPORT SERVICE*"[TITLE/ABSTRACT] OR "TALKING THERAP*"[TITLE/ABSTRACT] OR "MENTORING"[MESH TERMS] OR "COUNSELING"[MESH TERMS] OR "SOCIAL SUPPORT"[MESH TERMS]	1362940
4	1 AND 2 AND 3	1991
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3	TI ADVICE* OR TI ADVISE* OR TI ADVISING OR TI COACH* OR TI CONSULT* OR TI COUNSELLING OR TI COUNSELING OR TI INSTRUCT* OR TI GUIDANCE* OR TI GUIDE* OR TI MENTORING OR TI "PASTORAL CARE" OR TI "SOCIAL SUPPORT*" OR TI SUPERVIS* OR TI "SUPPORT SERVICE*" OR TI "TALKING THERAP*" OR AB ADVICE* OR AB ADVISE* OR AB ADVISING OR AB COACH* OR AB CONSULT* OR AB COUNSELLING OR AB COUNSELING OR AB INSTRUCT* OR AB GUIDANCE* OR AB GUIDE* OR AB MENTORING OR AB "PASTORAL CARE" OR AB "SOCIAL SUPPORT*" OR AB SUPERVIS* OR AB "SUPPORT SERVICE*" OR AB "TALKING THERAP*" OR MH COUNSELING+ OR MH PSYCHOSOCIAL SUPPORT+	575744
4	1 AND 2 AND 3	962
5	TI SURGICAL OR TI SURGERY OR TI GENETI* OR TI PROTEOMIC* OR TI GENOMIC* OR TI PROTEIN* OR TI NEUROIMAG* OR AB SURGICAL OR AB SURGERY OR AB GENETI* OR AB PROTEOMIC* OR AB GENOMIC* OR AB PROTEIN* OR AB NEUROIMAG*	643710
6	4 NOT 5	895

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4	1 AND 2 AND 3	802
5	(SURGICAL OR SURGERY OR GENETI* OR PROTEOMIC* OR GENOMIC* OR PROTEIN* OR NEUROIMAG*):TI,AB,KW	359548
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#	Entry	Hits
1	TS=(DEMENT* OR ALZHEIMER* OR "MAJOR NEUROCOGNITIVE DISORDER*") INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	343520
2	TS=(APP OR APPS OR "AUGMENTED REALIT*" OR "BLENDED TREATMENT" OR "BLENDED THERAPY" OR BLOG OR "CELL PHONE*" OR CELLPHONE* OR CHAT OR CHATS OR CHATBOT* OR COMPUTER* OR DIGITAL* OR "E COUNSELING" OR ECOUNSELING OR E-COUNSELING OR "E COUNSELLING" OR ECOUNSELLING OR E-COUNSELLING OR "E HEALTH" OR EHEALTH OR E-HEALTH OR ELECTRONIC* OR EMAIL* OR E-MAIL* OR FACEBOOK OR ICT OR INTERNET OR MESSENGER OR MHEALTH OR M-HEALTH OR "MOBILE DEVICE" OR "MOBILE HEALTH" OR "MOBILE PHONE" OR "MOBILE-HEALTH" OR ONLINE* OR PHONE OR REMOTE OR SKYPE OR SMARTPHONE* OR SMS OR "SOCIAL MEDIA" OR TABLET OR TABLETS OR TECHNOLOG* OR "TELE HOME CARE" OR "TELE MEDICINE" OR TELECARE OR TELE-CARE OR TELECARING OR TELE-CARING OR TELECOMMUNICATION OR TELECONFERENCE* OR TELE-CONFERENCE* OR TELECONSULTATION* OR TELE-CONSULTATION* OR TELEHEALTH OR TELE-HEALTH OR TELEMEDICINE OR TELENURSE OR TELENURSING OR TELEPHONE OR TELEPHONES OR TELEPRESENCE OR TELEPRESENT OR TELEREHABILITATION OR TELE-REHABILITATION OR TELETHERAP* OR TELE-THERAP* OR TELETREATMENT OR TELE-TREATMENT OR VIDEOCONFERENCE OR VIDEOCONFERENCES OR VIRTUAL OR "WEB BASED" OR WEBBASED OR WEB-BASED OR "WORLD WIDE WEB") INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	5975986
3	TS=(ADVICE* OR ADVISE* OR ADVISING OR COACH* OR CONSULT* OR COUNSELLING OR COUNSELING OR INSTRUCT* OR GUIDANCE* OR GUIDE* OR MENTORING OR "PASTORAL CARE" OR "SOCIAL SUPPORT*" OR SUPERVIS* OR "SUPPORT SERVICE*" OR "TALKING THERAP*") INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	2256839
4	1 AND 2 AND 3	2743
5	TS=(SURGICAL OR SURGERY OR GENETI* OR PROTEOMIC* OR GENOMIC* OR PROTEIN* OR NEUROIMAG*) INDEXES=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI TIMESPAN=ALL YEARS	7195603
6	4 NOT 5	2183

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Date: 20.04.2021

#	Entry	Hits
1	(DEMENT* OR ALZHEIMER* OR "MAJOR NEUROCOGNITIVE DISORDER*").TI,AB,MP. OR EXP DEMENTIA/ OR EXP ALZHEIMER DISEASE/	113751
2	(APP OR APPS OR "AUGMENTED REALIT*" OR "BLENDED TREATMENT" OR "BLENDED THERAPY" OR BLOG OR "CELL PHONE*" OR CELLPHONE* OR CHAT OR CHATS OR CHATBOT* OR COMPUTER* OR DIGITAL* OR "E COUNSELING" OR ECOUNSELING OR E-COUNSELING OR "E COUNSELLING" OR ECOUNSELLING OR E-COUNSELLING OR "E HEALTH" OR EHEALTH OR E-HEALTH OR ELECTRONIC* OR EMAIL* OR E-MAIL* OR FACEBOOK OR ICT OR INTERNET OR MESSENGER OR MHEALTH OR M-HEALTH OR "MOBILE DEVICE" OR "MOBILE HEALTH" OR "MOBILE PHONE" OR "MOBILE-HEALTH" OR ONLINE* OR PHONE OR REMOTE OR SKYPE OR SMARTPHONE* OR SMS OR "SOCIAL MEDIA" OR TABLET OR TABLETS OR TECHNOLOG* OR "TELE HOME CARE" OR "TELE MEDICINE" OR TELECARE OR TELE-CARE OR TELECARING OR TELE-CARING OR TELECOMMUNICATION OR TELECONFERENCE* OR TELE-CONFERENCE* OR TELECONSULTATION* OR TELE-CONSULTATION* OR TELEHEALTH OR TELE-HEALTH OR TELEMEDICINE OR TELENURSE OR TELENURSING OR TELEPHONE OR TELEPHONES OR TELEPRESENCE OR TELEPRESENT OR TELEREHABILITATION OR TELE-REHABILITATION OR TELETHERAP* OR TELE-THERAP* OR TELETREATMENT OR TELE-TREATMENT OR VIDEOCONFERENCE OR VIDEOCONFERENCES OR VIRTUAL OR "WEB BASED" OR WEBBASED OR WEB-BASED OR "WORLD WIDE WEB").TI,AB,MP. OR EXP TELEMEDICINE/ OR EXP TECHNOLOGY/ OR EXP VIRTUAL REALITY/ OR EXP TELECONFERENCING/ OR EXP ONLINE THERAPY/ OR EXP TELEREHABILITATION/ OR EXP SOCIAL MEDIA/ OR EXP MOBILE DEVICES/	506844
3	(ADVICE* OR ADVISE* OR ADVISING OR COACH* OR CONSULT* OR COUNSELLING OR COUNSELING OR INSTRUCT* OR GUIDANCE* OR GUIDE* OR MENTORING OR "PASTORAL CARE" OR "SOCIAL SUPPORT*" OR SUPERVIS* OR "SUPPORT SERVICE*" OR "TALKING THERAP*").TI,AB,MP. OR EXP COUNSELING/ OR EXP SOCIAL SUPPORT/	700791
4	1 AND 2 AND 3	1151
5	(SURGICAL OR SURGERY OR GENETI* OR PROTEOMIC* OR GENOMIC* OR PROTEIN* OR NEUROIMAG*).TI,AB,MP.	285132
6	4 NOT 5	994

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	16
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Na
Support:			
Sources	5a	Indicate sources of financial or other support for the review	16
Sponsor	5b	Provide name for the review funder and/or sponsor	16
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	16
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4ff.
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7ff.
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	9f.
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	10f.
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	10f.
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	11
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	11
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	11
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	11
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	11ff.
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	11ff.
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	11ff.
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	11ff.
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	14
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	14

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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